

DSC 411/511
Advance Data Mining Algorithms and Applications

Fall 2026

Instructor: Ijaz Ul Haq, Ph.D.

ijazhaq@umich.edu

Location: MSB 104



Lecture 05
Classification
(Decision trees, Naive Bayes)

Previous class:

- ***Principle component analysis (PCA)***
  - ***Where to use it?***
  - ***How it works?***

- ***Features selection***
  - Filter
  - Embedded

- ***Apply PCA and feature selection on AMES dataset***

Today's class:

- ***What is classification?***
- ***Decision tress***
- ***Naïve Bayes***
- ***Related NoteBook (code)***

Where We Left Off Last Class?

Four Lectures of Data Preparation

- You classified every column by type.
- You found the problems and fixed them.
- You encoded, scaled, and reduced.
- You now have data an algorithm can actually read.

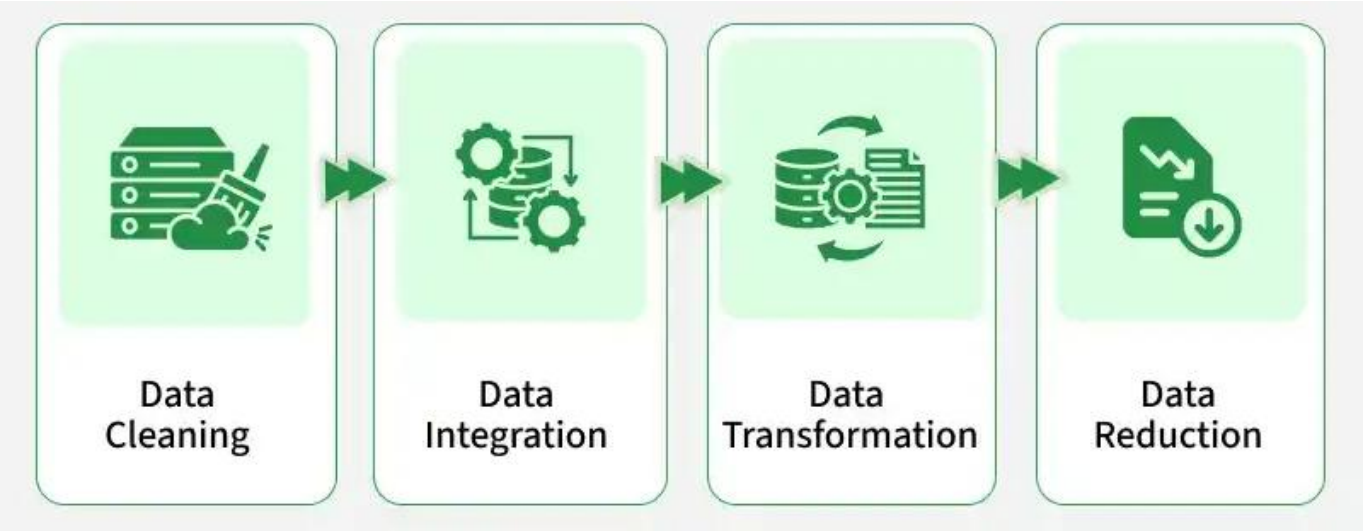


Image source: https://www.geeksforgeeks.org/data-science/data-preprocessing-in-data-mining/

Today we finally use it to predict something.

Classification

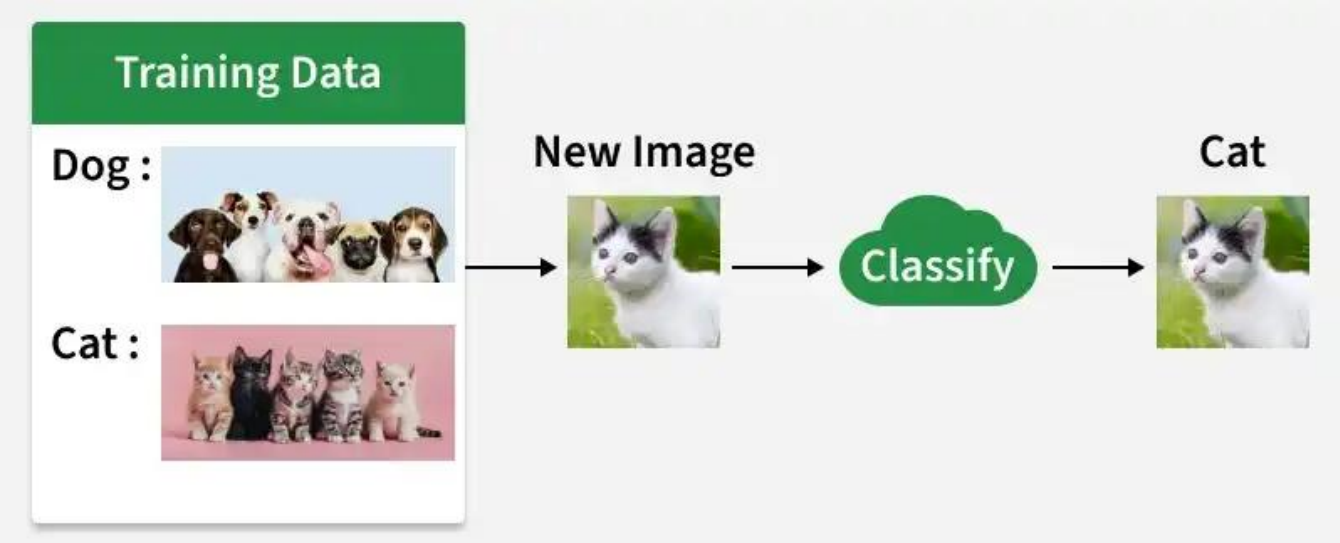
Definition:

Classification is the task of assigning objects to one of several predefined categories.

Chapter 3
Introduction to Data Mining, 2nd Edition

What that means in practice:

- You have examples that are already labelled.
- The model learns the pattern that connects the columns to the label.
- Then it labels new rows you have never seen.



Classification

Types of classification:

- **Binary classification:**
  Two possible classes, and each row gets exactly one. i.e., Spam or not spam. Above or below the median price.

- **Multiclass classification**
  More than two classes, and each row still gets exactly one. i.e., Which of 28 neighbourhoods is this house in?

- **Multi-label classification**
  More than two classes, and a row can have several at once. A news article tagged both Politics and Economics.

The two methods we cover today

Method	The idea in one line
Decision Tree	Ask a series of yes/no questions until you reach an answer
Naive Bayes	Work out which class is most probable, given the evidence

Classification is **supervised**.
The training data is already labelled, and that is what you learn from.

Classification: Decision Trees

Definition:

A decision tree is a hierarchical structure of nodes and directed edges, where each internal node tests one attribute, each branch is an outcome of that test, and each leaf holds a class label.

Chapter 3
Introduction to Data Mining, 2nd Edition

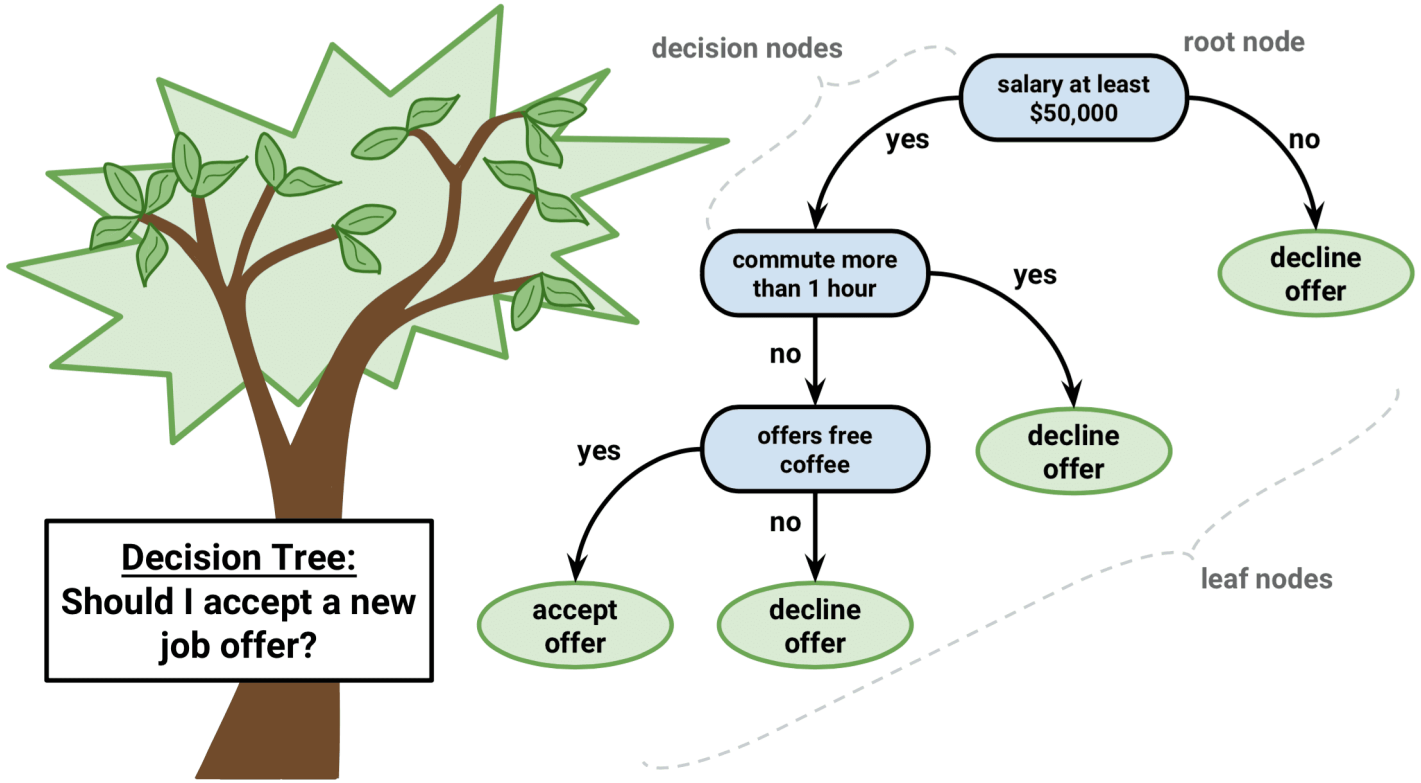


Image source: https://dataaspirant.com/how-decision-tree-algorithm-works/

Classification: Decision Trees

Where it can be used?

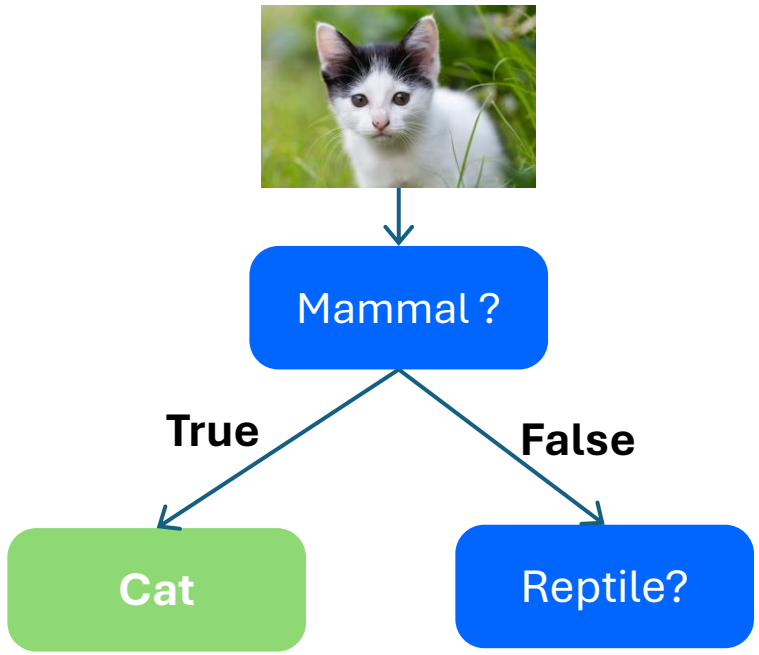
- **Credit approval.** The bank must give a reason for a refusal. A tree is a set of reasons.
- **Medical triage.** Nurses follow decision trees on paper every day.
- **Customer churn.** Output is a rule marketing can act on, not a score.

How it works?

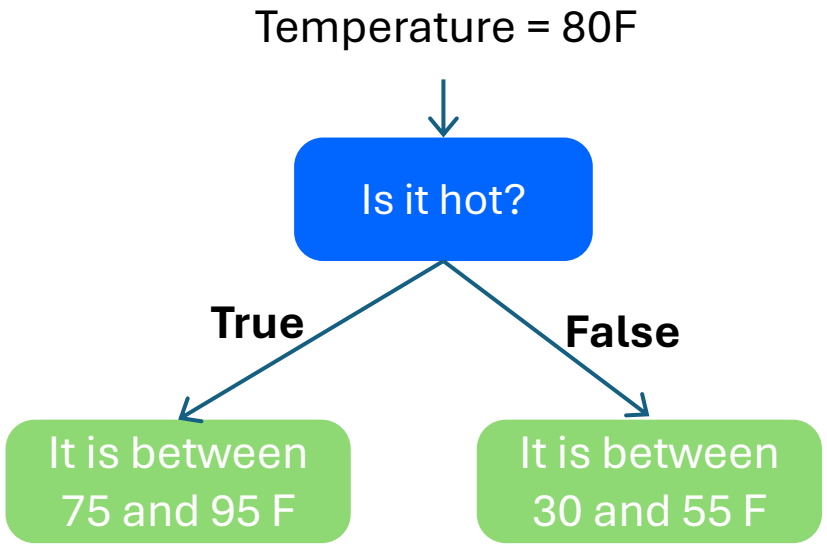
1. Look at every column and every possible **split point**.
2. Score each split by **how well** it separates the classes.
3. Take the best one. Divide the data into two branches.
4. Repeat on each branch.
5. Stop when a rule says stop: maximum depth, too few rows, or a **pure node**.
6. To predict, walk a new row down the tree until it reaches a **leaf**.

Classification: Decision Trees

The Working of Decision Trees?



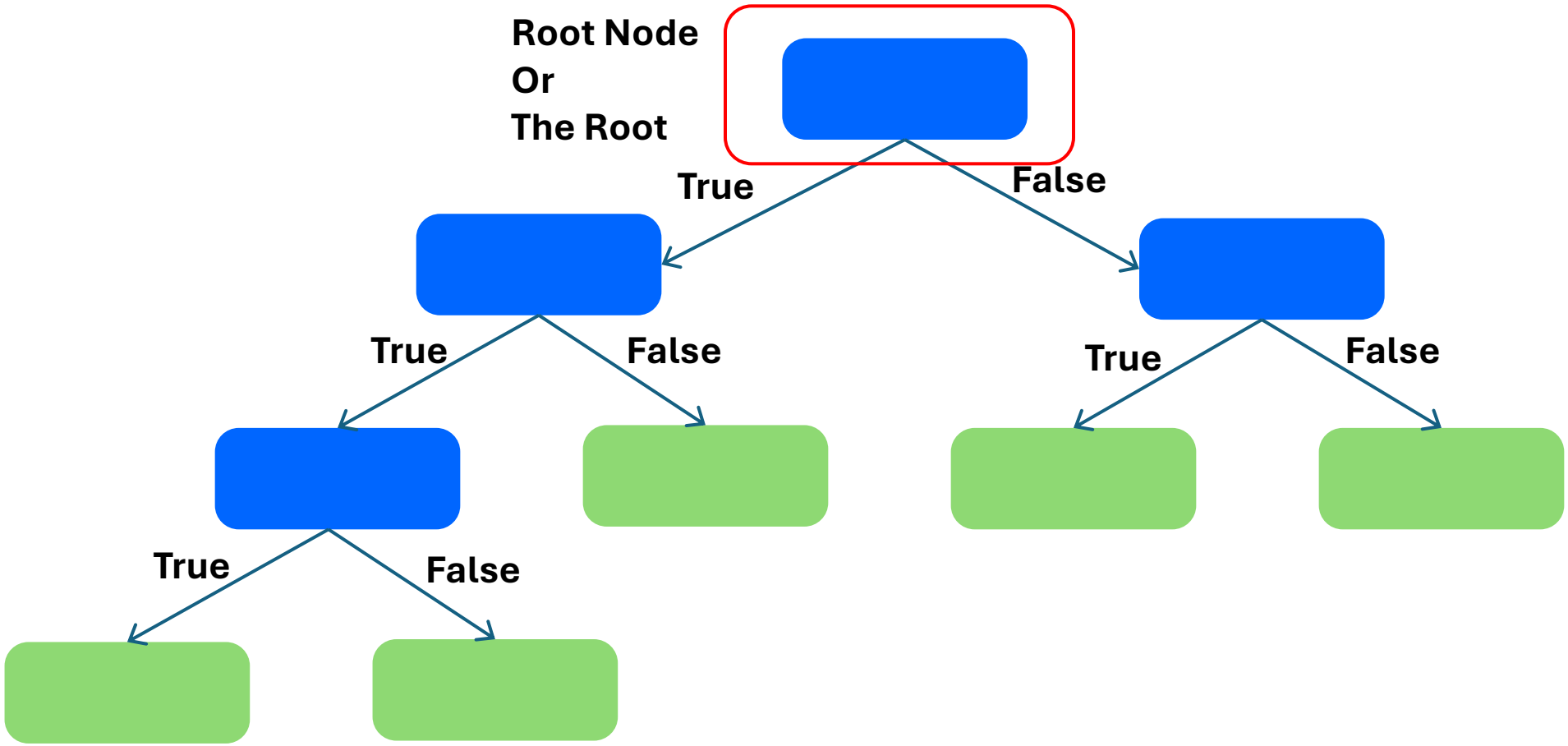
Classification Tree
if it classify things into categories



Regression Tree
if it classify things into numeric values

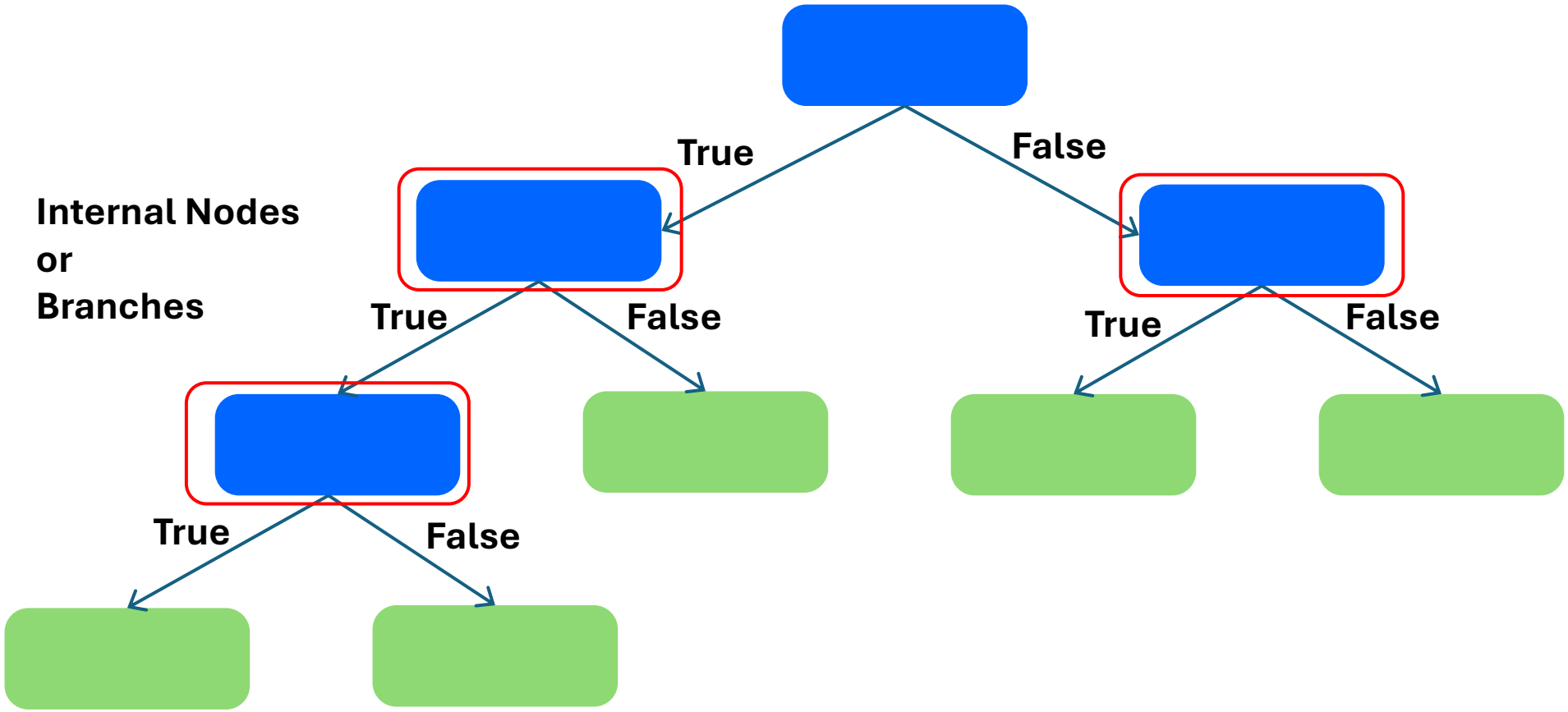
Classification: Decision Trees

The Working of Decision Trees?



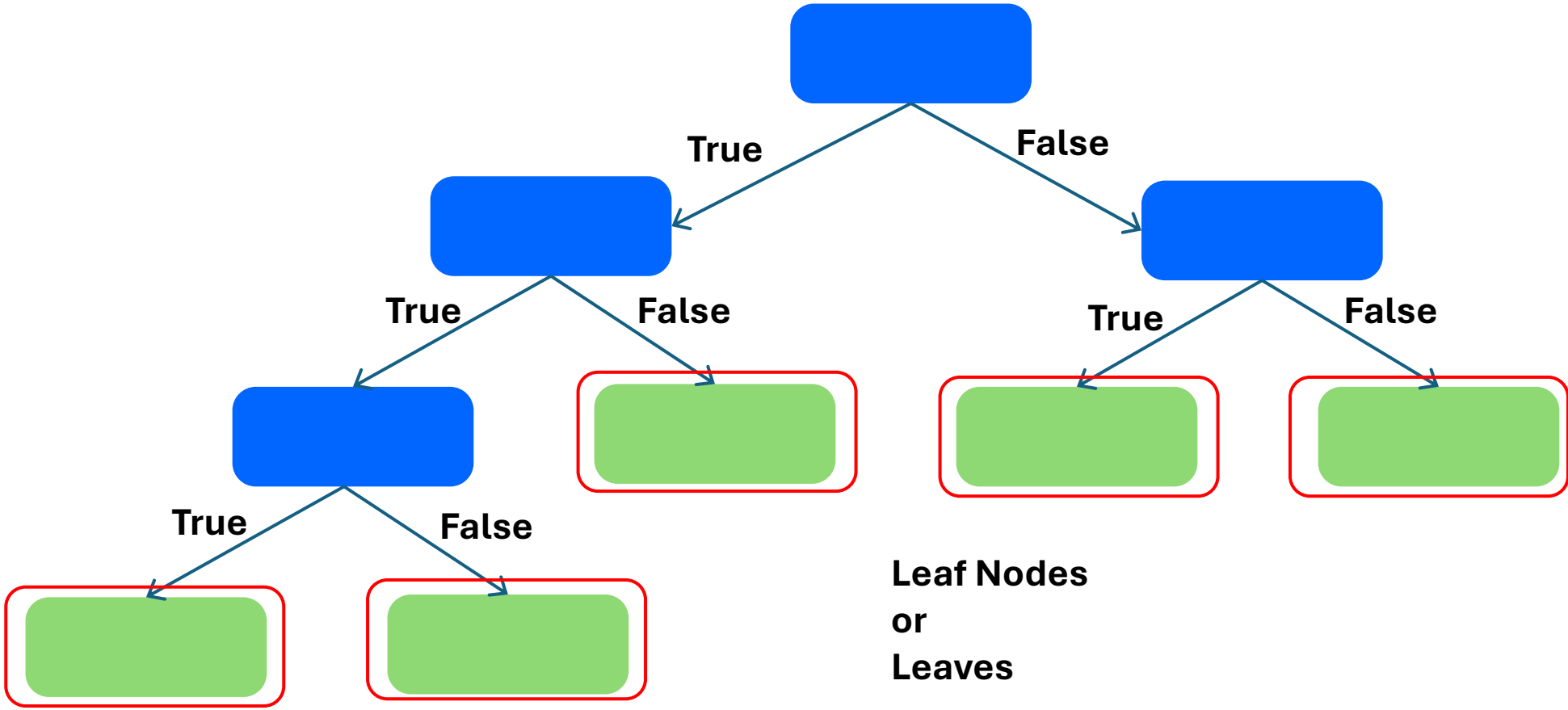
Classification: Decision Trees

The Working of Decision Trees?



Classification: Decision Trees

The Working of Decision Trees?



Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

Classification: Decision Trees

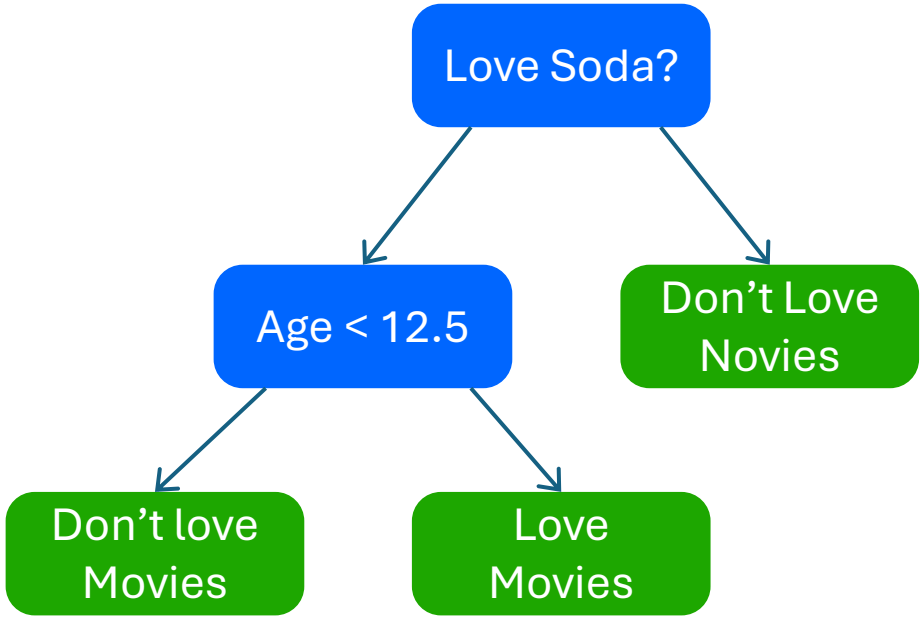
The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

Classification: Decision Trees

The Working of Decision Trees?

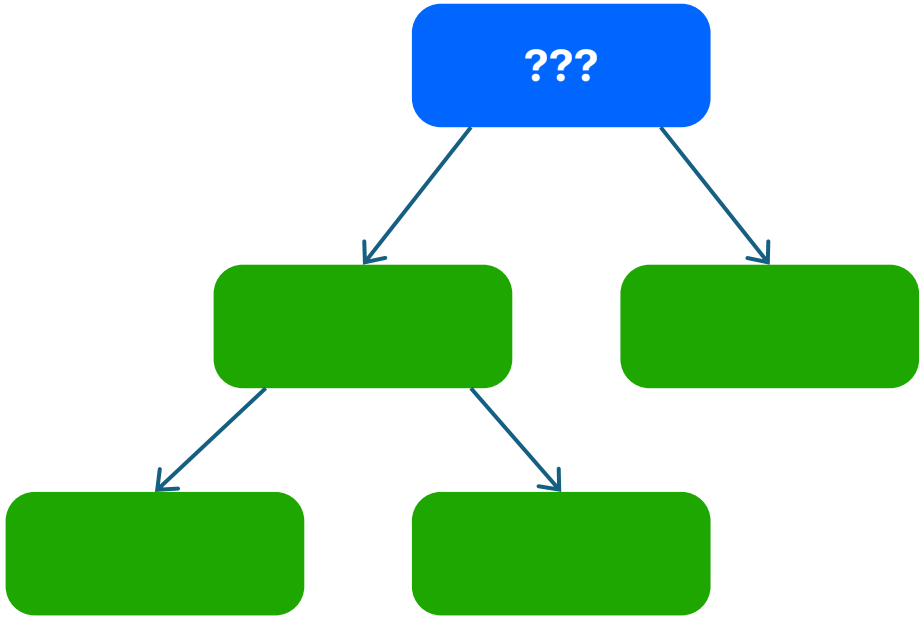
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Classification: Decision Trees

The Working of Decision Trees?

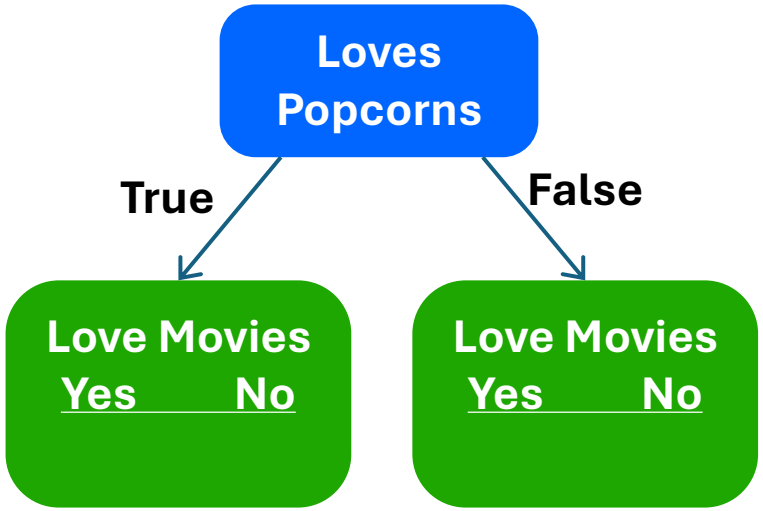
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Classification: Decision Trees

The Working of Decision Trees?

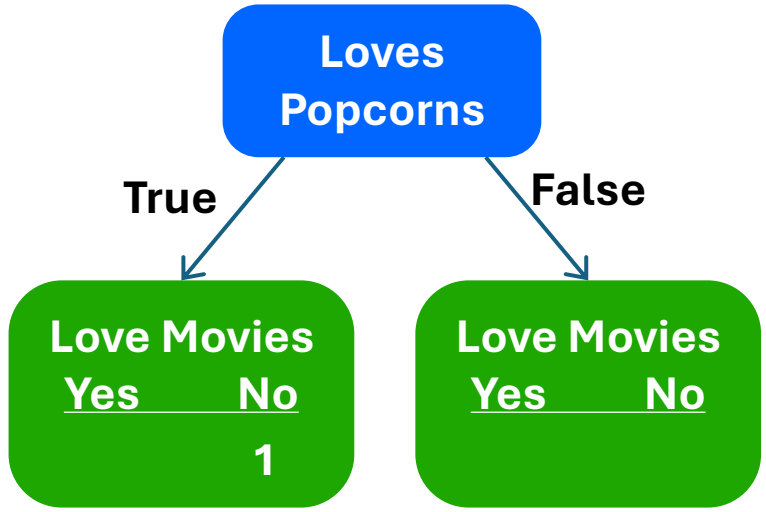
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Classification: Decision Trees

The Working of Decision Trees?

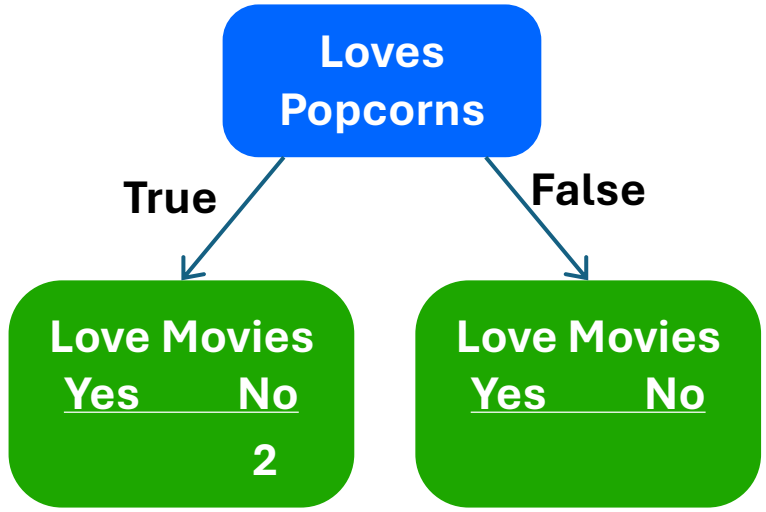
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Classification: Decision Trees

The Working of Decision Trees?

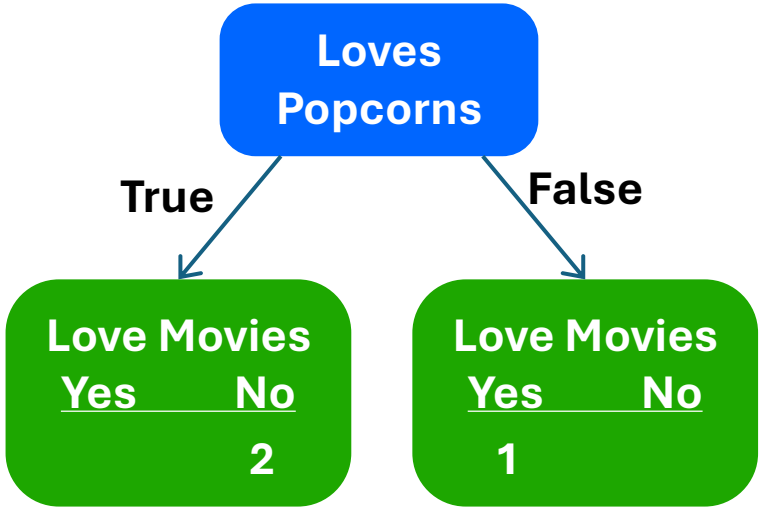
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Classification: Decision Trees

The Working of Decision Trees?

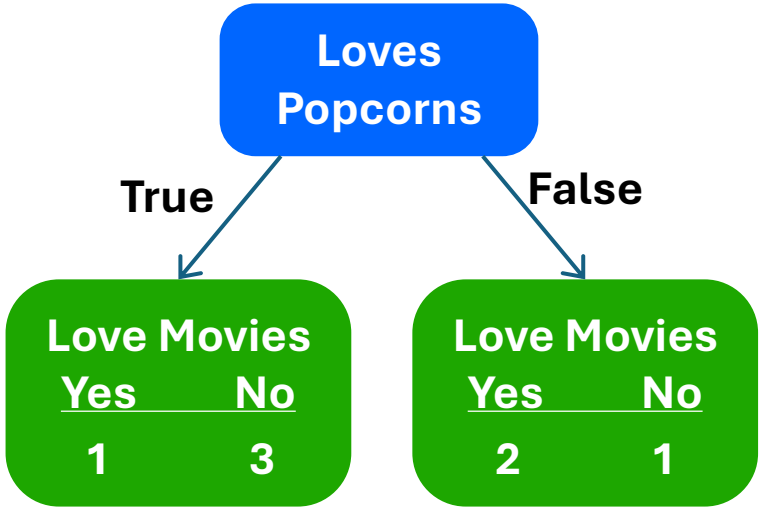
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Classification: Decision Trees

The Working of Decision Trees?

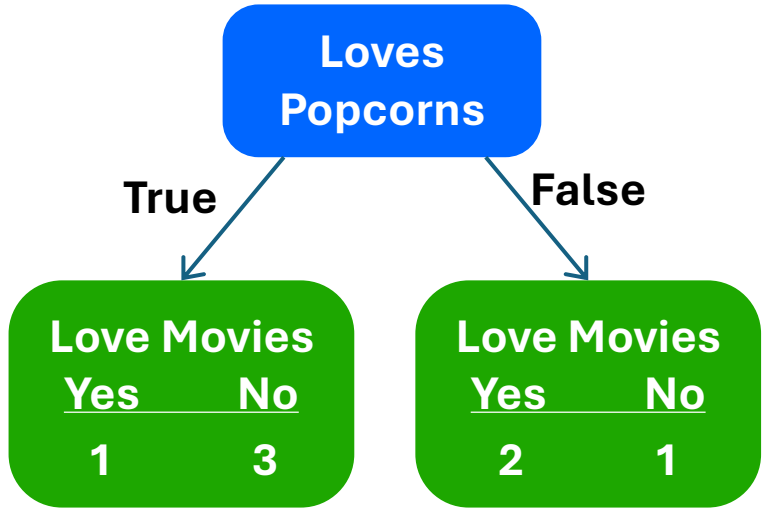
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

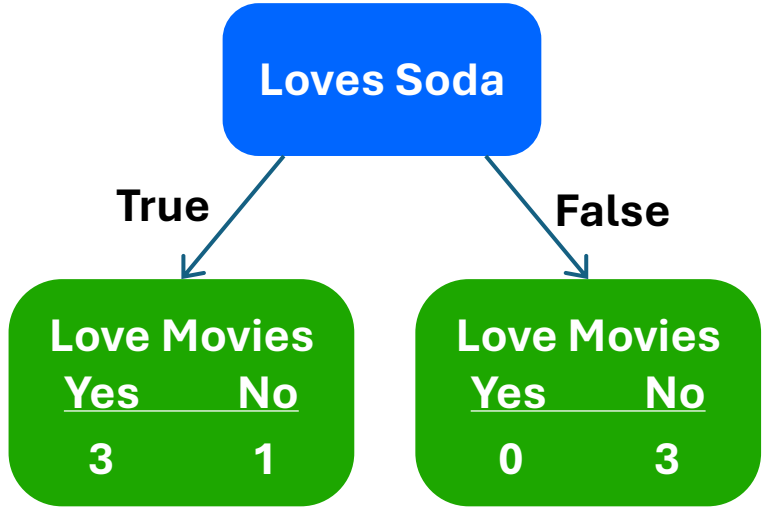


Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

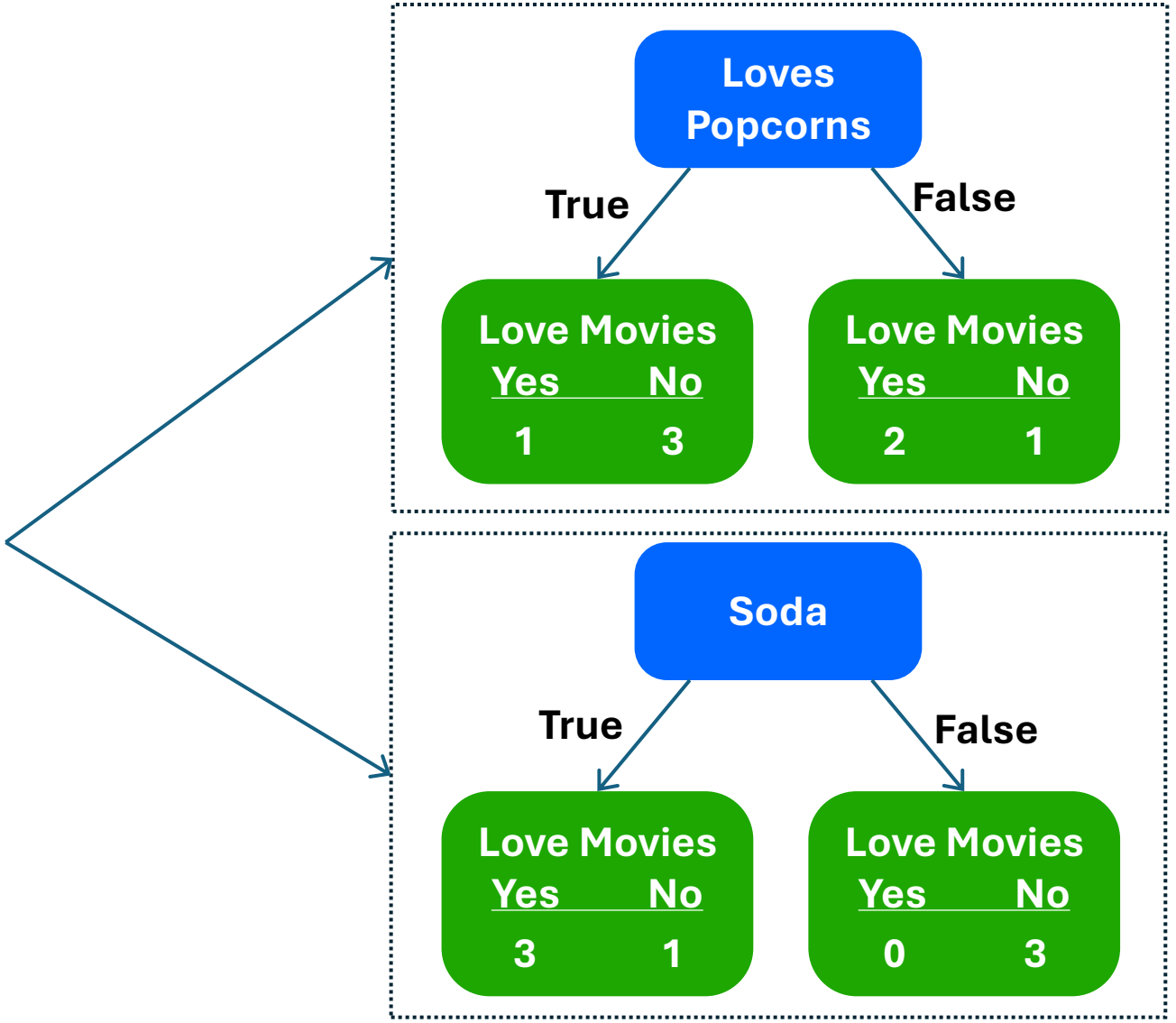




Classification: Decision Trees

The Working of Decision Trees?

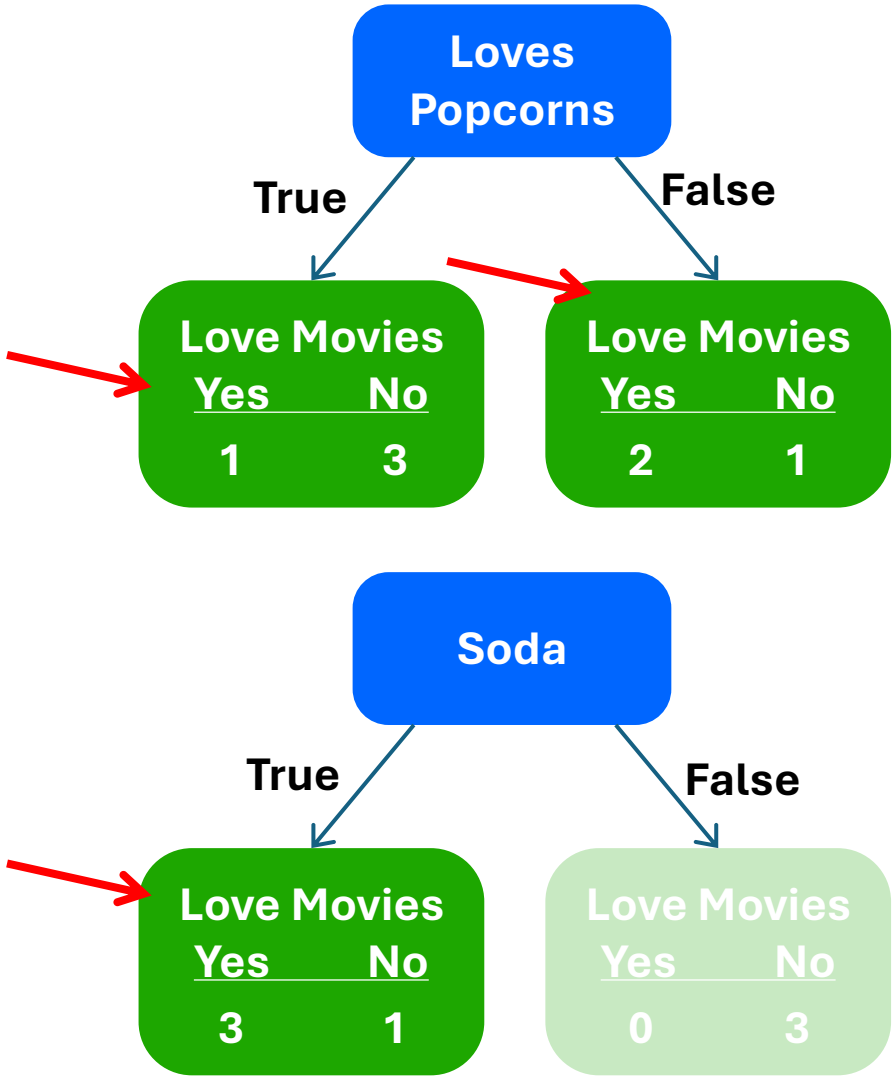
Neither one of the two little trees does the perfect job predicting who **loves movies**



Classification: Decision Trees

The Working of Decision Trees?

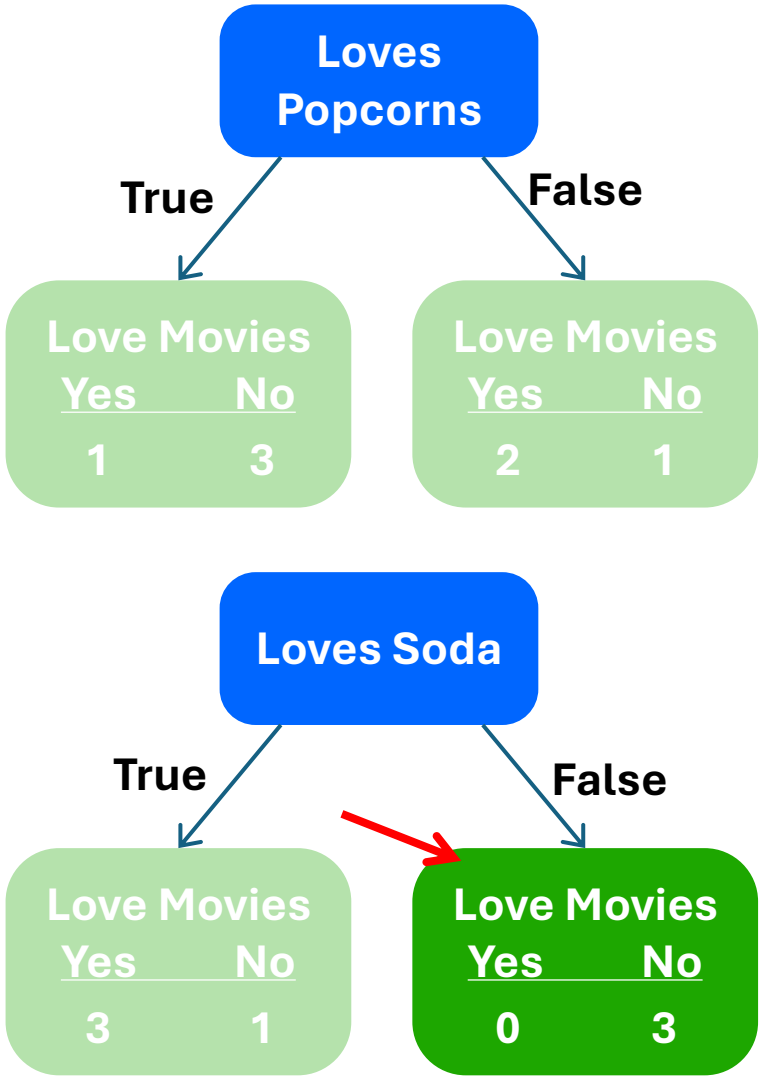
Mixture of people in these leaves, who **do** and **do not love movies,** They are called **Impure**



Classification: Decision Trees

The Working of Decision Trees?

In contrast this Leaf only contains people who **do not love movies**



Classification: Decision Trees

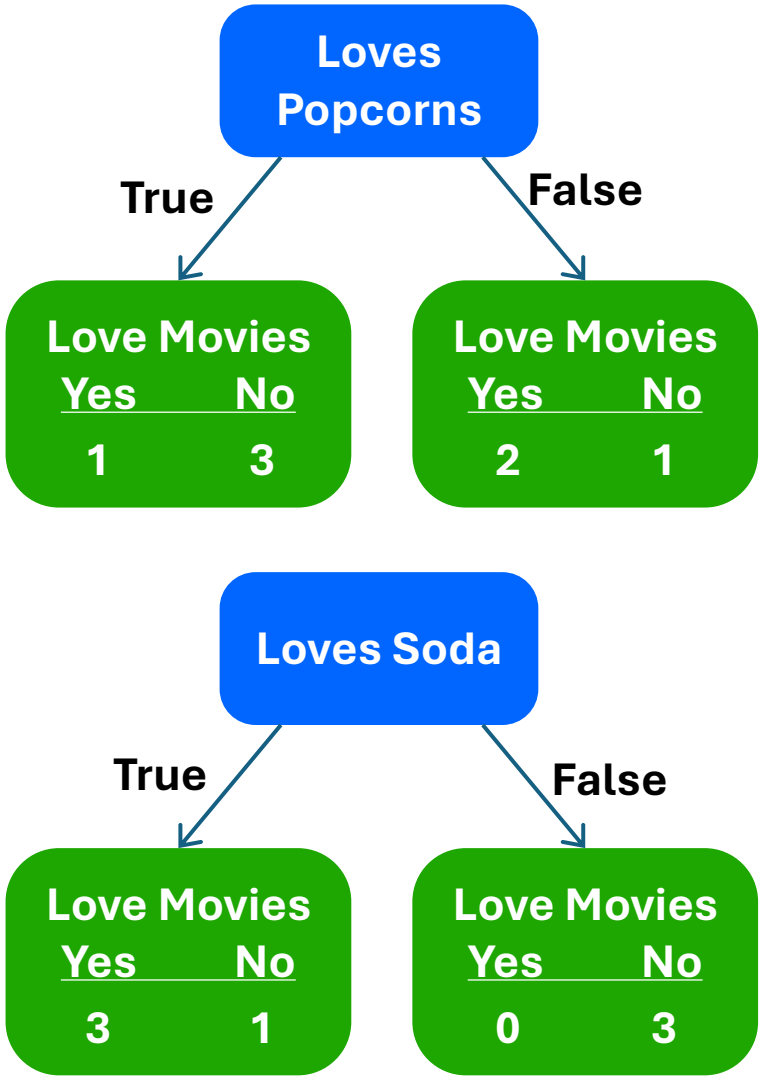
The Working of Decision Trees?

Which tree will do a better job predicting who will love and will not love movies?

Loves popcorn have 2 impure leaves, and **Loves Soda** have only one, so it seems **Loves soda** does better job than **loves popcorn**

Can we quantify the impurity of leaves?

Yes, there are several methods to find it but for simplicity we will go with **Gini Impurity.**

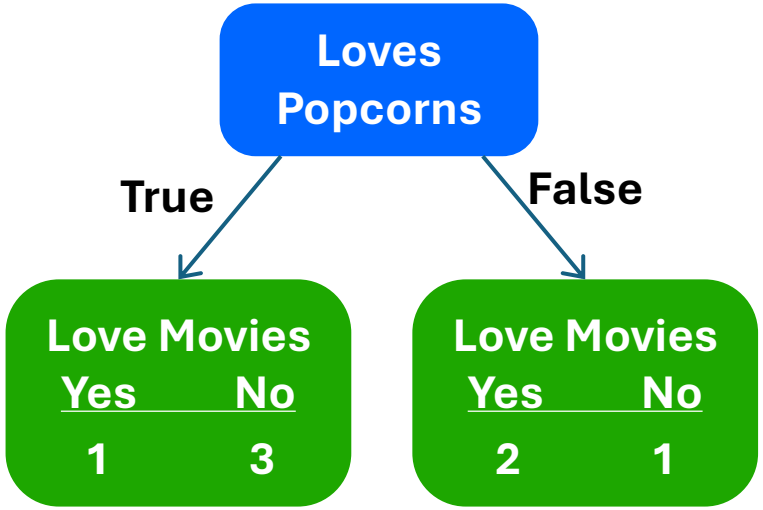


Classification: Decision Trees

The Working of Decision Trees?

Gini Impurity: 1 minus the sum of the squared probabilities for each class in a dataset or node

Gini Impurity Formula: $$G = 1 - \sum_{i=1}^{C} p_i^2$$

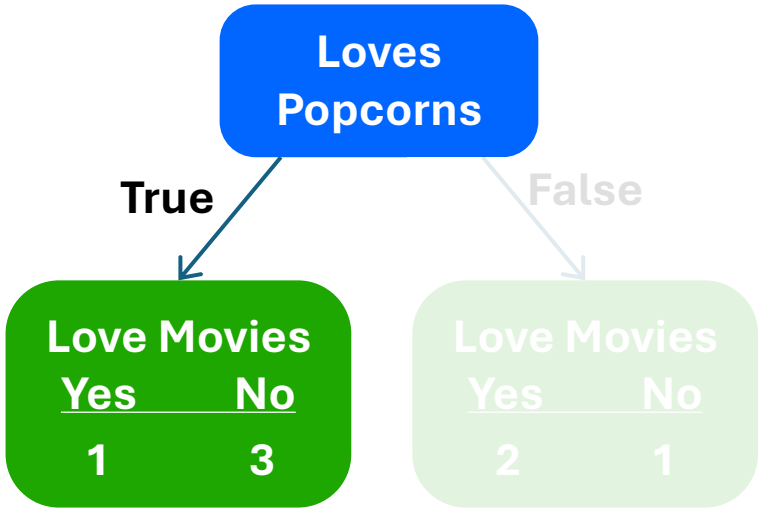


Classification: Decision Trees

The Working of Decision Trees?

Gini Impurity: 1 minus the sum of the squared probabilities for each class in a dataset or node

Gini Impurity Formula: $$G = 1 - \sum_{i=1}^{C} p_i^2$$



Gini Impurity for a leaf : 1 – (probability of "yes")^2 - (probability of "No")^2

Gini Impurity for a leaf : $$1 - \left(\frac{1}{1+3}\right)^2 - \left(\frac{3}{1+3}\right)^2$$

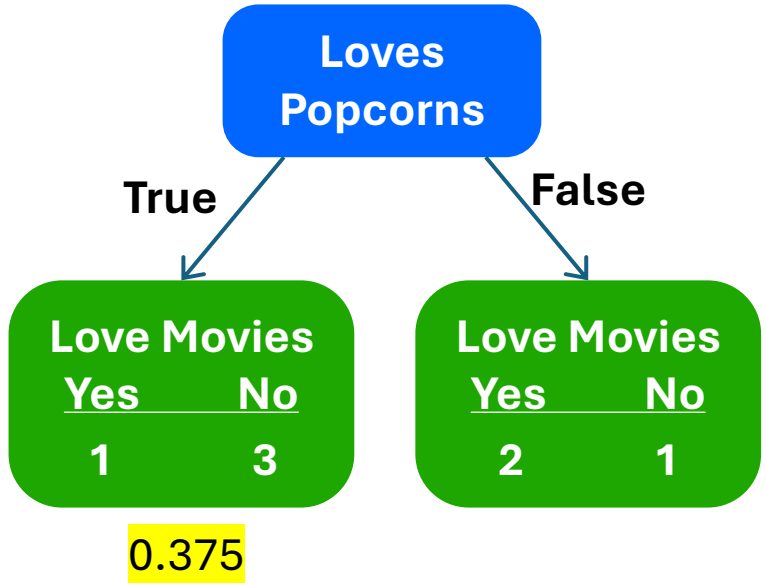
Gini Impurity for a leaf : 0.375

Classification: Decision Trees

The Working of Decision Trees?

Gini Impurity: 1 minus the sum of the squared probabilities for each class in a dataset or node

Gini Impurity Formula: $$G = 1 - \sum_{i=1}^{C} p_i^2$$



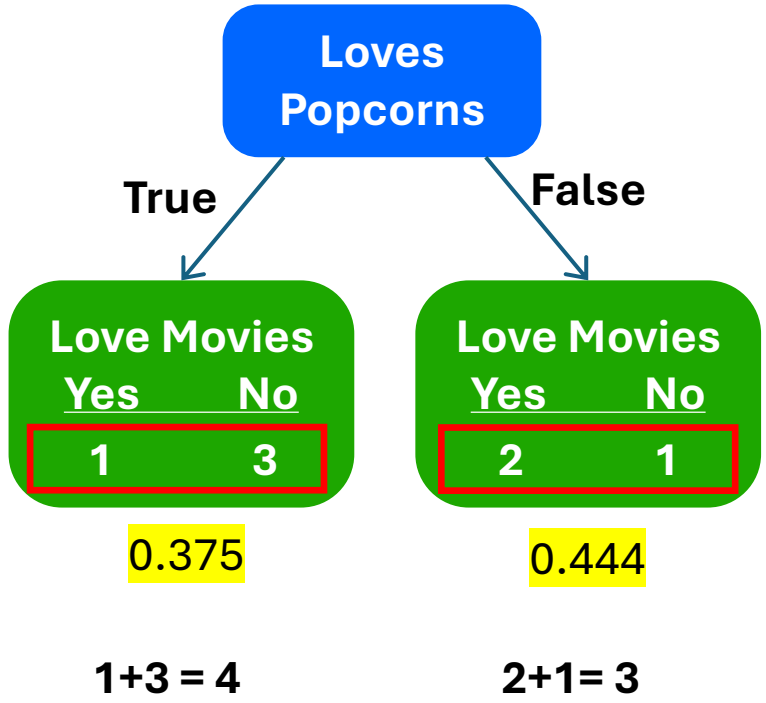
Gini Impurity for a leaf : 1 – (probability of "yes")^2 - (probability of "No")^2

Gini Impurity for a leaf : $$1 - \left(\frac{2}{2+1}\right)^2 - \left(\frac{1}{2+1}\right)^2$$

Gini Impurity for a leaf : 0.444

Classification: Decision Trees

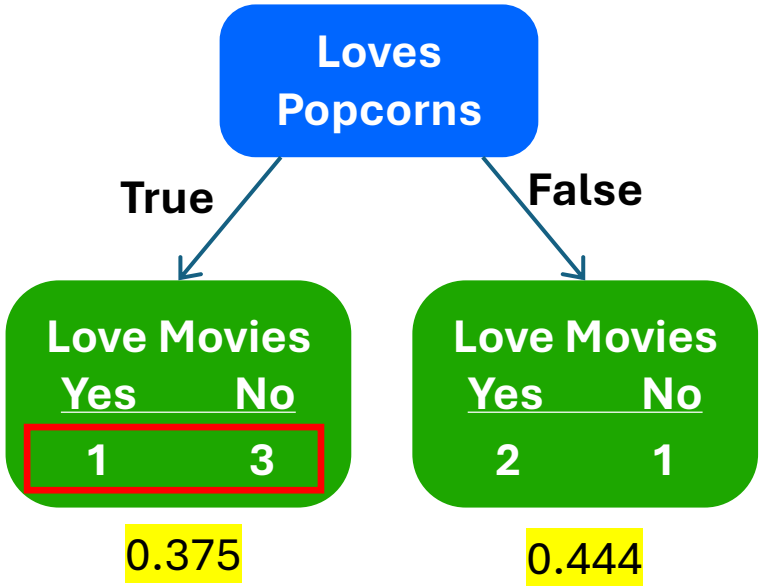
The Working of Decision Trees?



The leaves do not represent the same number of people. We need total Impurity!

Classification: Decision Trees

The Working of Decision Trees?



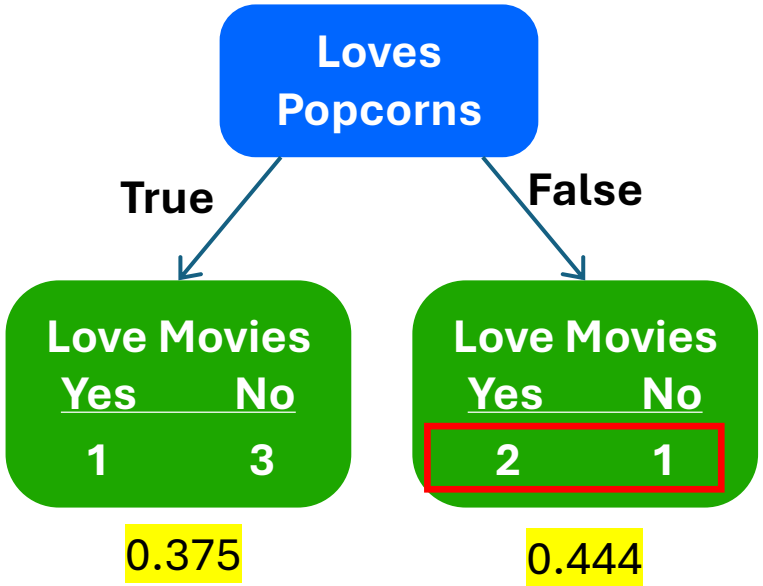
Gini Impurity for a leaf : weighted avg of **Gini impurities** for **leaves**

Gini Impurity for a leaf : $\left(\frac{\text{total ppl in left leaf}}{\text{total ppl in both leaves}}\right)$ x Gini impurity

Gini Impurity for a leaf : $\left(\frac{4}{4+3}\right)$ x 0.375

Classification: Decision Trees

The Working of Decision Trees?



Gini Impurity for a leaf : weighted avg of **Gini impurities** for **leaves**

Gini Impurity for a leaf : $\left(\frac{\text{total ppl in left leaf}}{\text{total ppl in both leaves}}\right)$ x Gini impurity

Gini Impurity for a leaf : $\left(\frac{4}{4+3}\right)$ x 0.375 + $\left(\frac{3}{4+3}\right)$ x 0.444

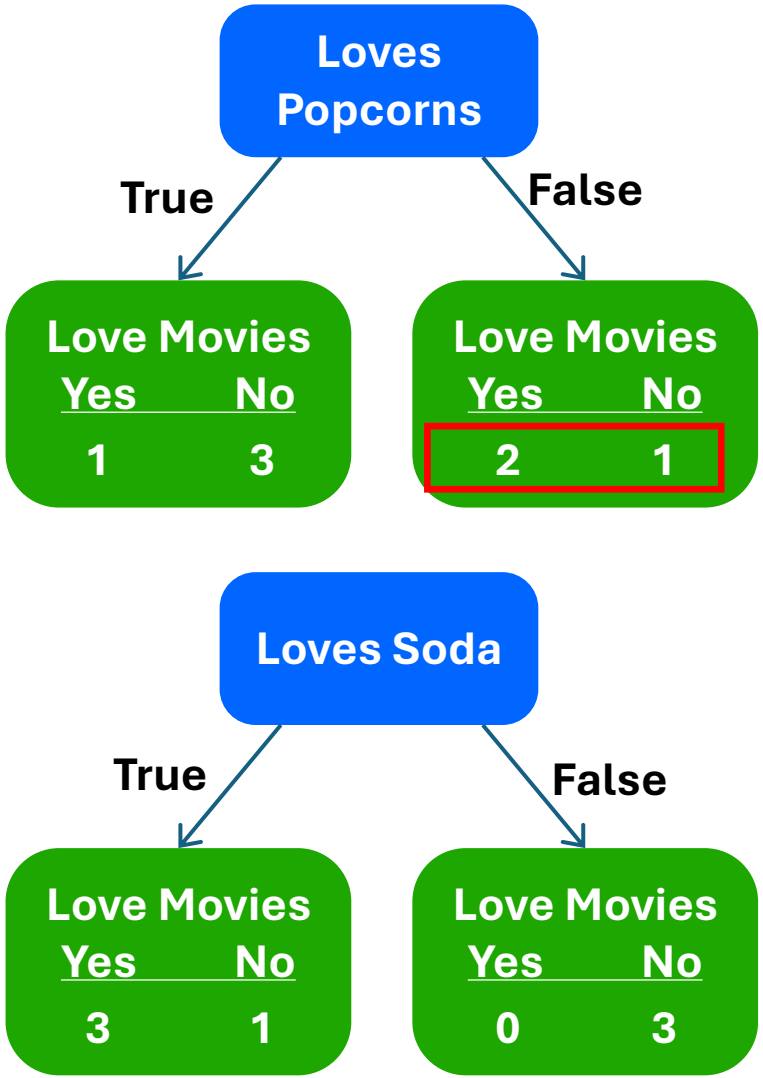
Gini Impurity for love popcorn: 0.405

Classification: Decision Trees

The Working of Decision Trees?

Gini Impurity for a love popcorn : 0.405

Gini Impurity for a love soda : 0.214

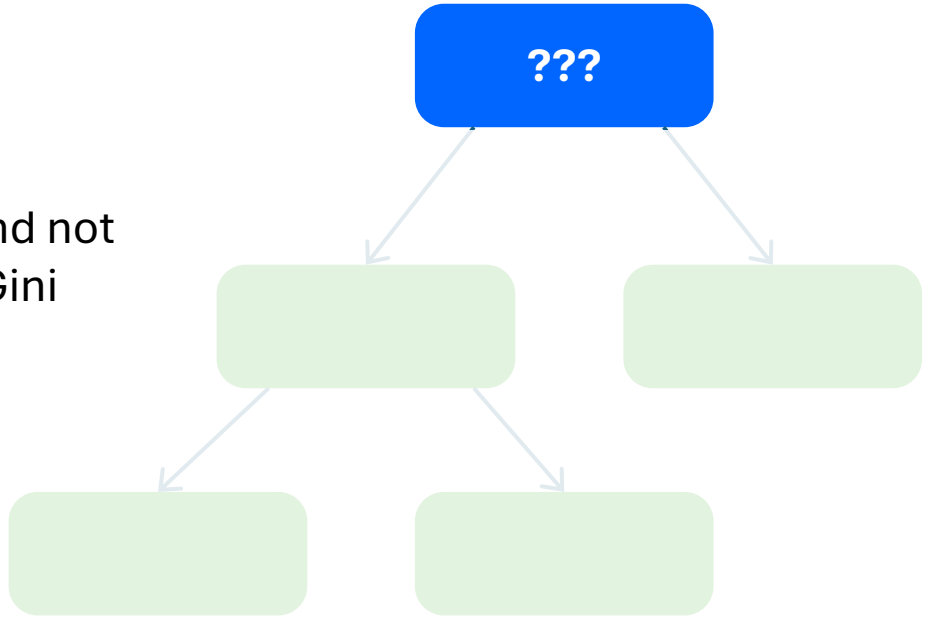


Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No
0.405	0.214		

However, age is numeric data and not just yes/no values, calculating Gini impurity is little more involved.



Classification: Decision Trees

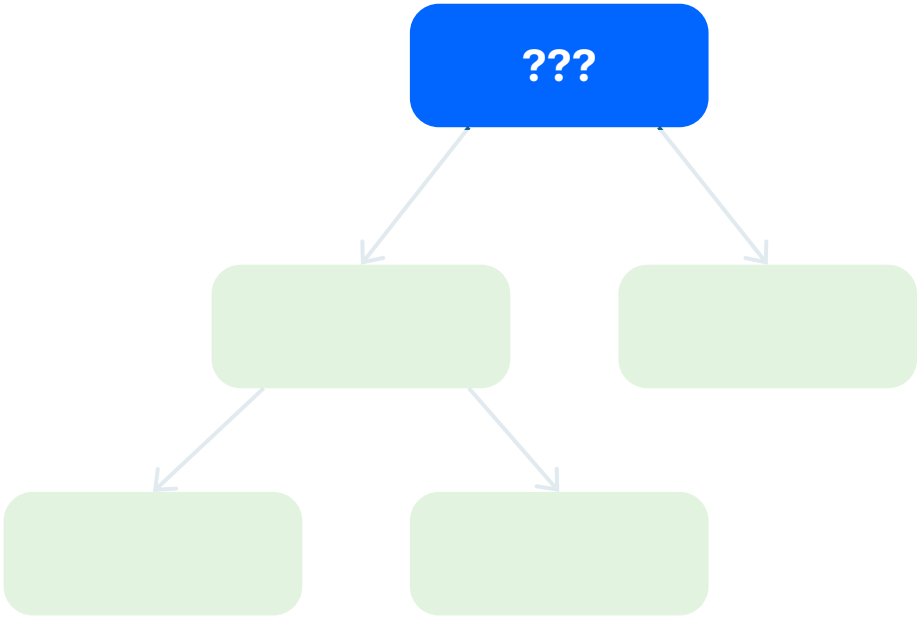
The Working of Decision Trees?

Age	Love Movies
7	No
12	No
18	Yes
35	Yes
38	Yes
50	No
83	No

9.5

Step1: First sort the rows by age, from lowest to the highest.

Step2: calculate the avg for all adjacent people.



Classification: Decision Trees

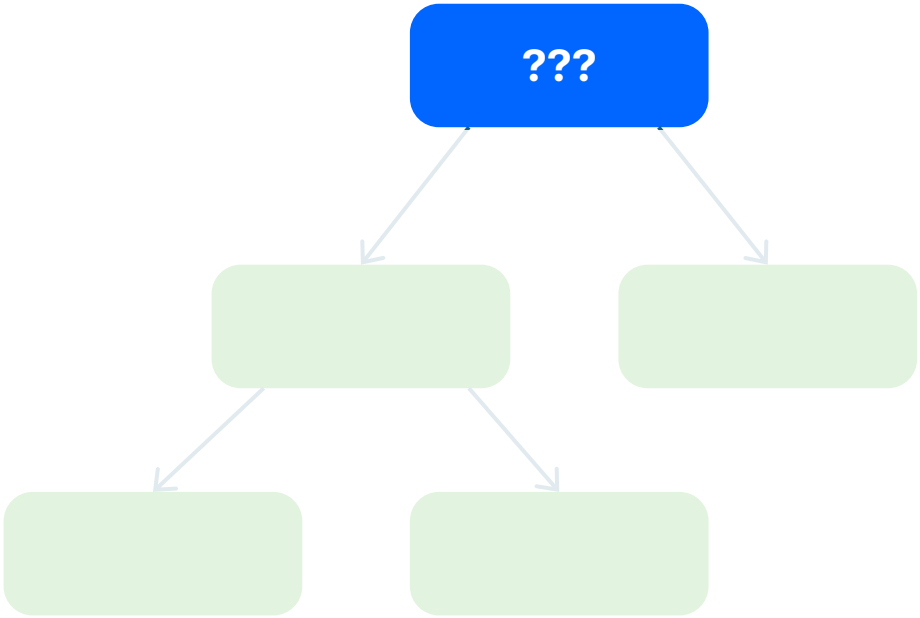
The Working of Decision Trees?

Age	Love Movies
7	No
12	No
18	Yes
35	Yes
38	Yes
50	No
83	No

9.5

Step1: First sort the rows by age, from lowest to the highest.

Step2: calculate the avg for all adjacent people.



Classification: Decision Trees

The Working of Decision Trees?

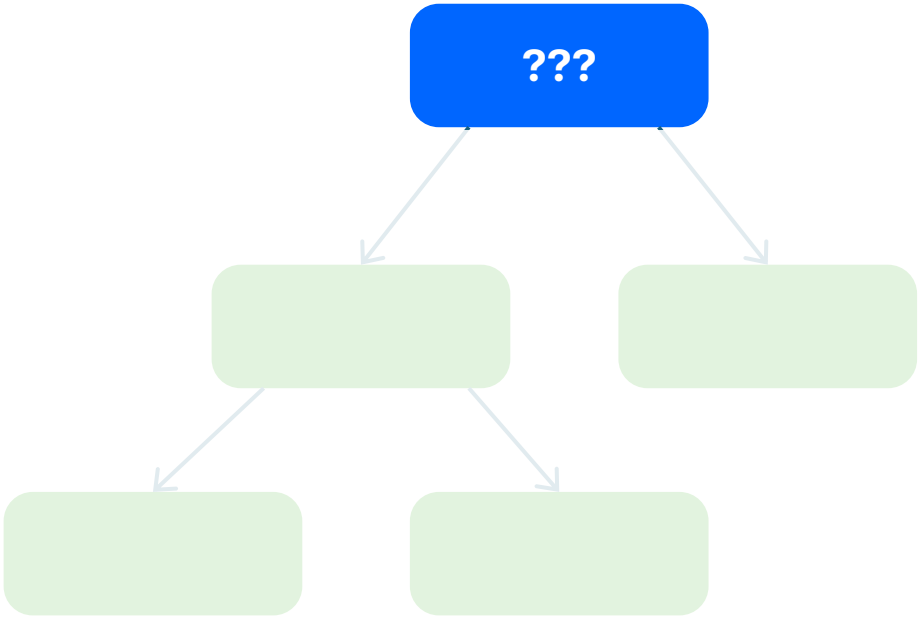
Age	Love Movies
7	No
12	No
18	Yes
35	Yes
38	Yes
50	No
83	No

9.5

15

Step1: First sort the rows by age, from lowest to the highest.

Step2: calculate the avg for all adjacent people.



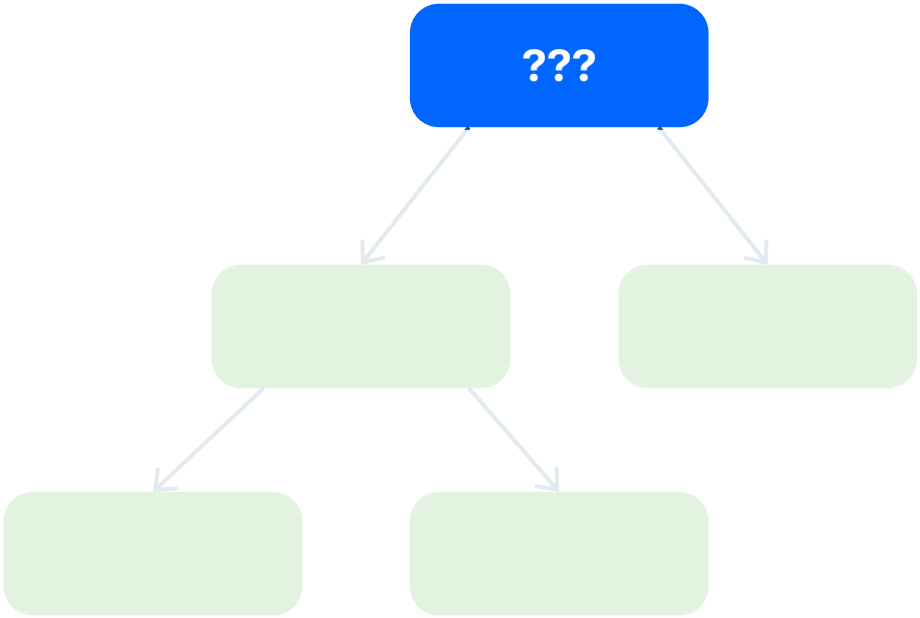
Classification: Decision Trees

The Working of Decision Trees?

	Age	Love Movies
	7	No
9.5 ←		
	12	No
15 ←		
	18	Yes
26.5 ←		
	35	Yes
	38	Yes
	50	No
	83	No

Step1: First sort the rows by age, from lowest to the highest.

Step2: calculate the avg for all adjacent people.



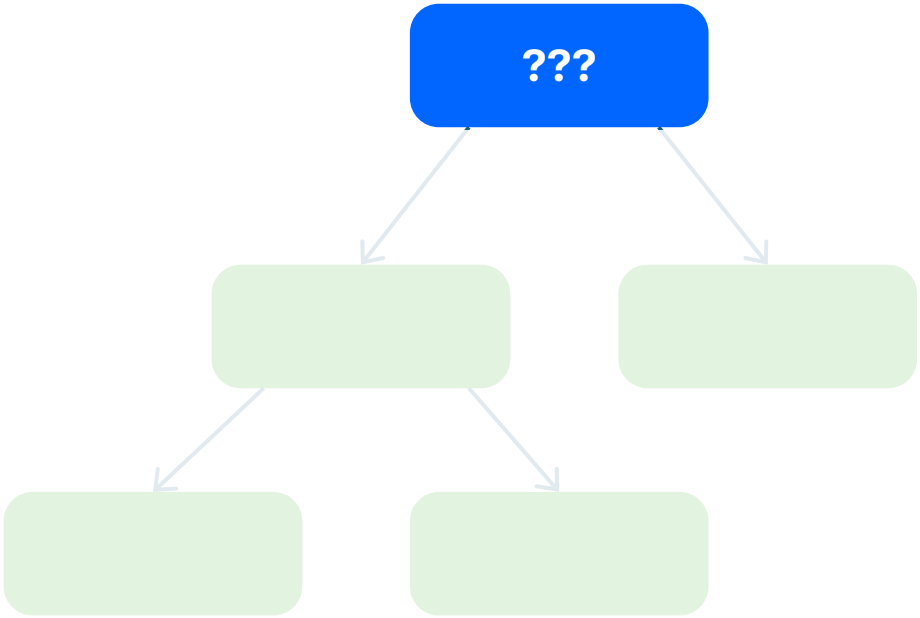
Classification: Decision Trees

The Working of Decision Trees?

	Age	Love Movies
	7	No
9.5 ←		
	12	No
15 ←		
	18	Yes
26.5 ←		
	35	Yes
36.5 ←		
	38	Yes
44 ←		
	50	No
66.5 ←		
	83	No

Step1: First sort the rows by age, from lowest to the highest.

Step2: calculate the avg for all adjacent people.



Classification: Decision Trees

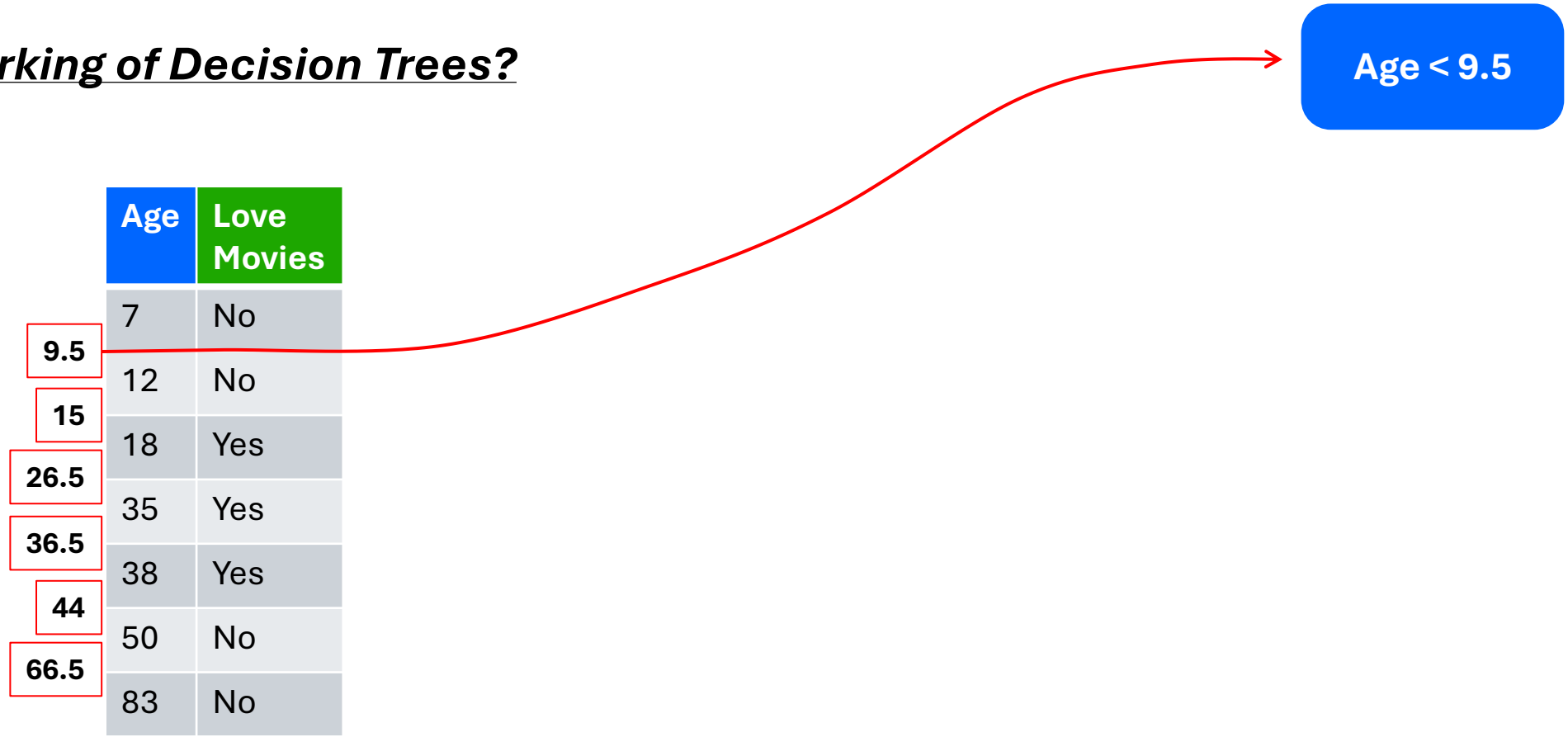
The Working of Decision Trees?

	Age	Love Movies
9.5	7	No
	12	No
15	18	Yes
26.5	35	Yes
36.5	38	Yes
44	50	No
66.5	83	No

→ Gini Impurity ?

Classification: Decision Trees

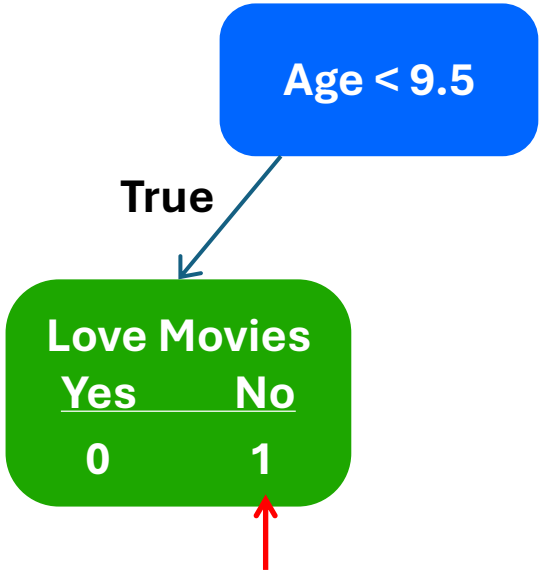
The Working of Decision Trees?



Classification: Decision Trees

The Working of Decision Trees?

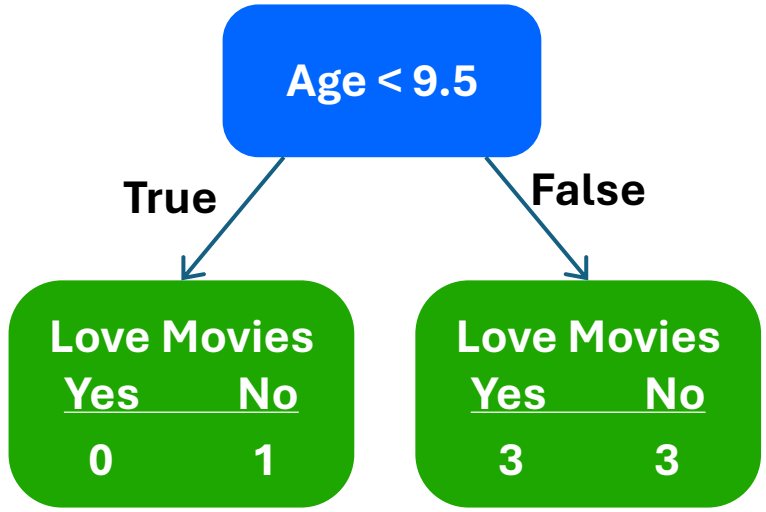
	Age	Love Movies
9.5	7	No
15	12	No
26.5	18	Yes
36.5	35	Yes
44	38	Yes
66.5	50	No
	83	No



Classification: Decision Trees

The Working of Decision Trees?

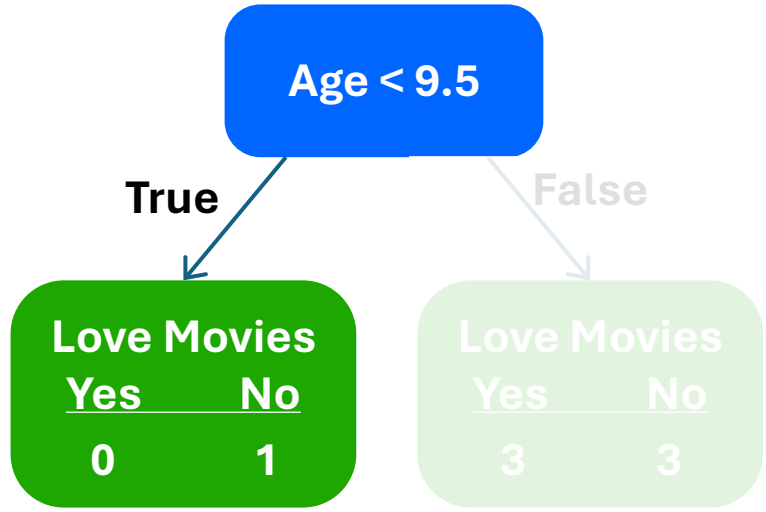
	Age	Love Movies
	7	No
9.5	12	No
15	18	Yes
26.5	35	Yes
36.5	38	Yes
44	50	No
66.5	83	No



Classification: Decision Trees

The Working of Decision Trees?

	Age	Love Movies
9.5	7	No
15	12	No
26.5	18	Yes
36.5	35	Yes
44	38	Yes
66.5	50	No
	83	No



Gini Impurity = 0

Gini Impurity for a leaf : 1 – (probability of "yes")^2 - (probability of "No")^2

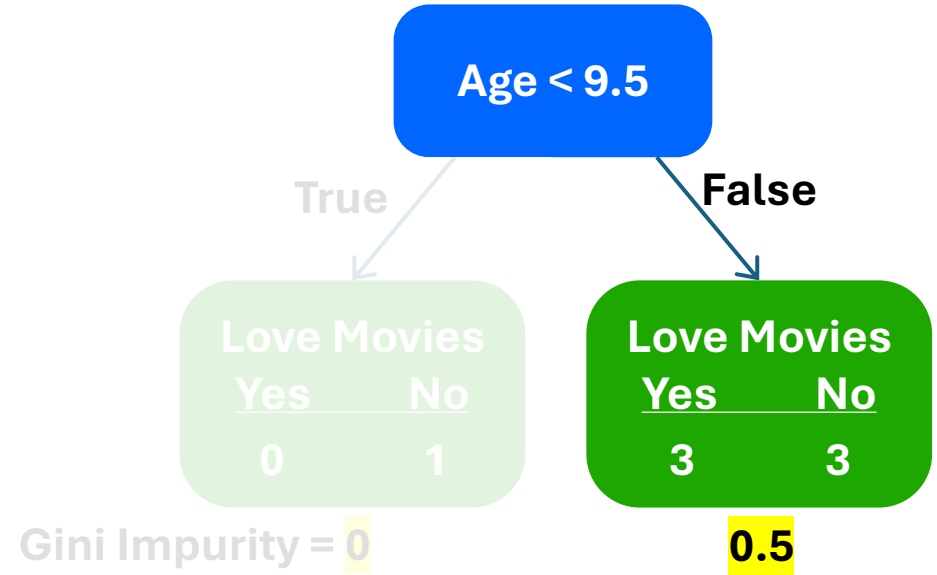
Gini Impurity for a leaf : $1 - \left(\frac{0}{0+1}\right)^2 - \left(\frac{1}{0+1}\right)^2$

Gini Impurity for a leaf : 0

Classification: Decision Trees

The Working of Decision Trees?

	Age	Love Movies
	7	No
9.5		
	12	No
15		
	18	Yes
26.5		
	35	Yes
36.5		
	38	Yes
44		
	50	No
66.5		
	83	No



Gini Impurity for a leaf : 1 – (probability of "yes")^2 - (probability of "No")^2

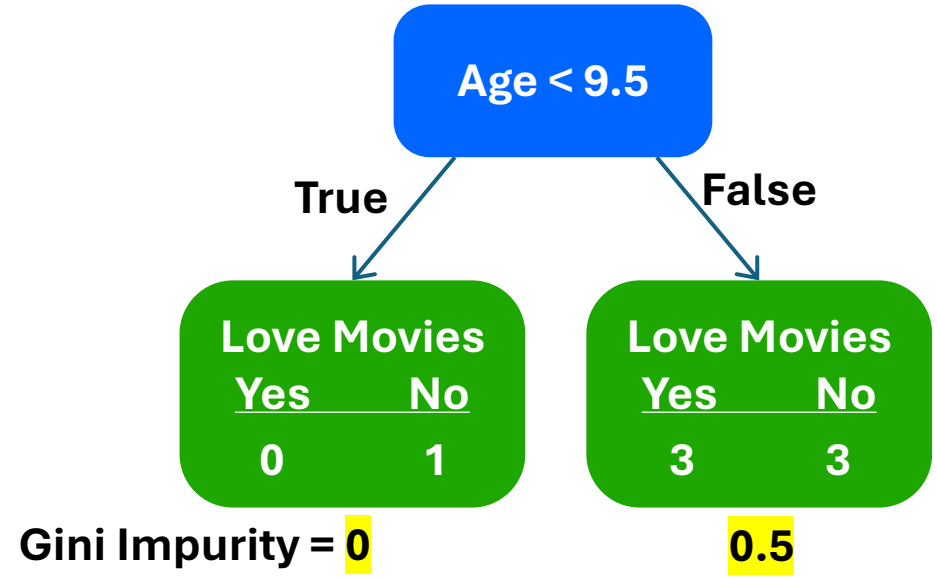
Gini Impurity for a leaf : $1 - \left(\frac{3}{3+3}\right)^2 - \left(\frac{3}{3+3}\right)^2$

Gini Impurity for a leaf : 0.5

Classification: Decision Trees

The Working of Decision Trees?

	Age	Love Movies
	7	No
9.5		
	12	No
15		
	18	Yes
26.5		
	35	Yes
36.5		
	38	Yes
44		
	50	No
66.5		
	83	No

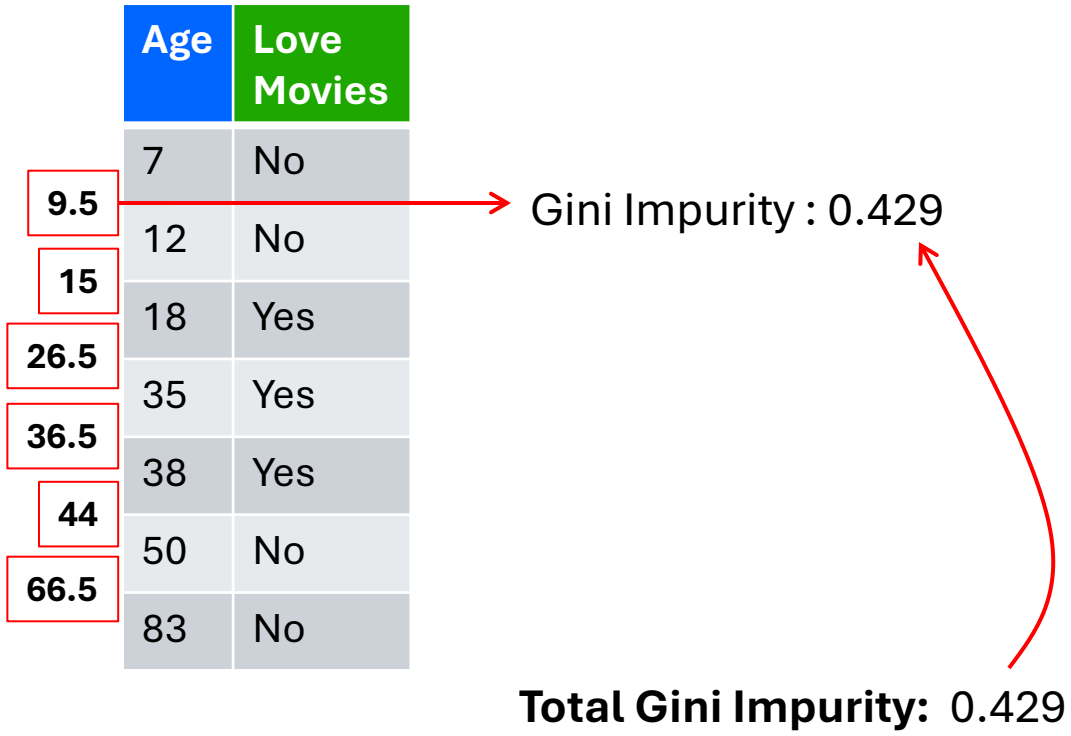


Total Gini Impurity: $\left(\frac{1}{1+6}\right)$ X 0 + $\left(\frac{6}{1+6}\right)$ X 0.5

Total Gini Impurity: 0.429

Classification: Decision Trees

The Working of Decision Trees?



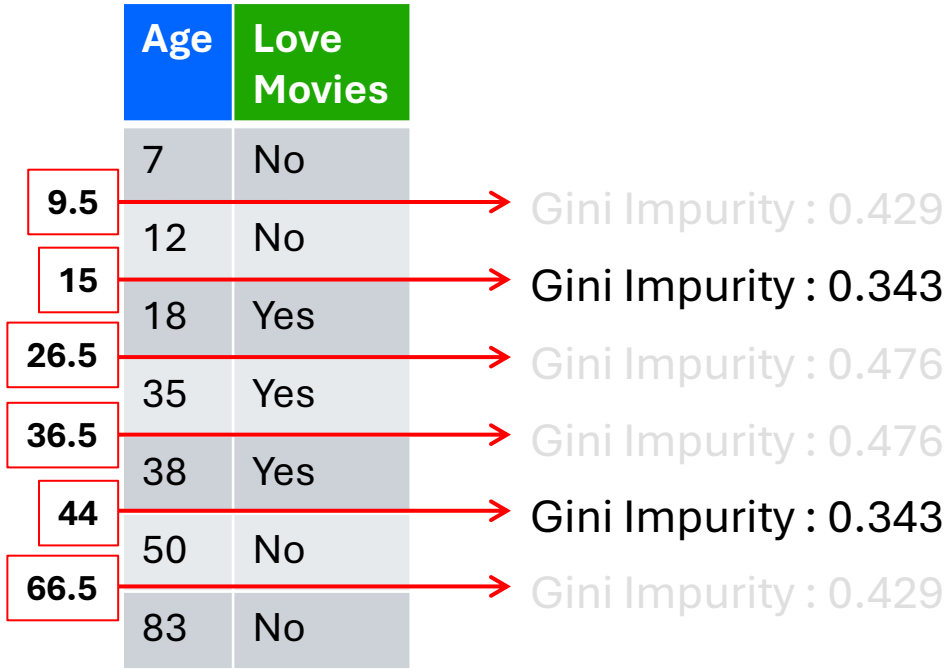
Classification: Decision Trees

The Working of Decision Trees?

Age	Love Movies	
7	No	
		9.5 → Gini Impurity : 0.429
12	No	
		15 → Gini Impurity : 0.343
18	Yes	
		26.5 → Gini Impurity : 0.476
35	Yes	
		36.5 → Gini Impurity : 0.476
38	Yes	
		44 → Gini Impurity : 0.343
50	No	
		66.5 → Gini Impurity : 0.429
83	No	

Classification: Decision Trees

The Working of Decision Trees?



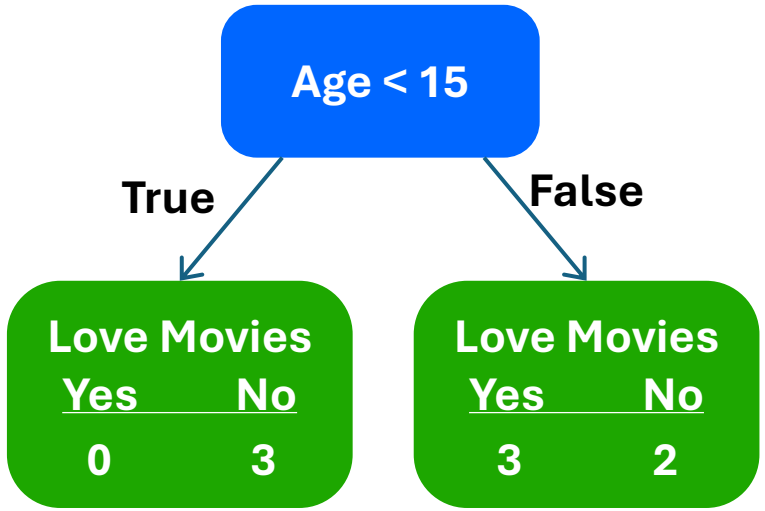
These two candidate thresholds, **15** and **44,** are tied for the lowest **Impurity, 0.343**

Classification: Decision Trees

The Working of Decision Trees?

	Age	Love Movies
9.5	7	No
15	12	No
26.5	18	Yes
36.5	35	Yes
44	38	Yes
66.5	50	No
	83	No

Gini Impurity : **0.343**



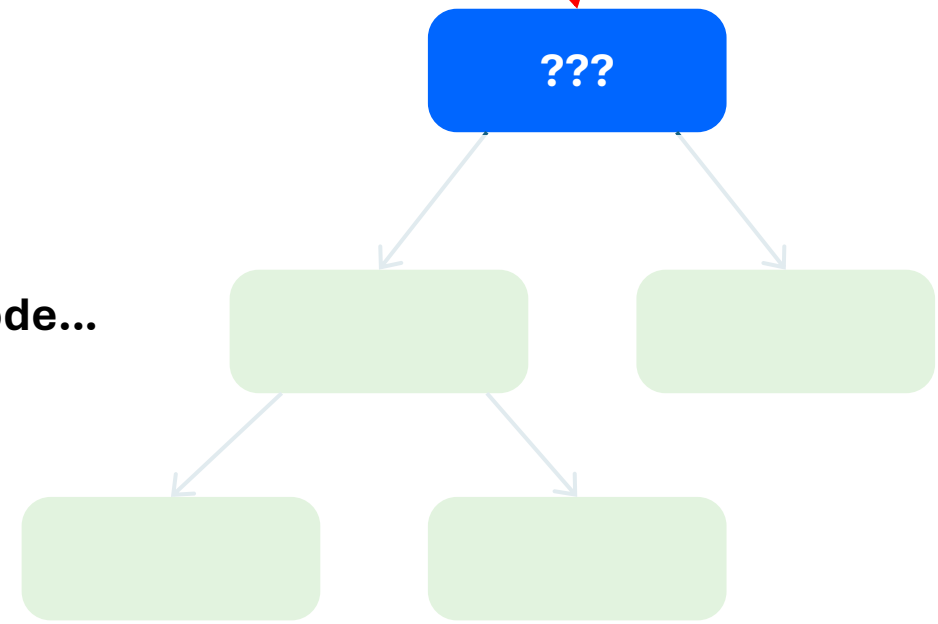
Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

However, remember that we are comparing **Gini Impurity** values for all three, Age, loves popcorn, loves soda,

For what? **To decide the root node…**



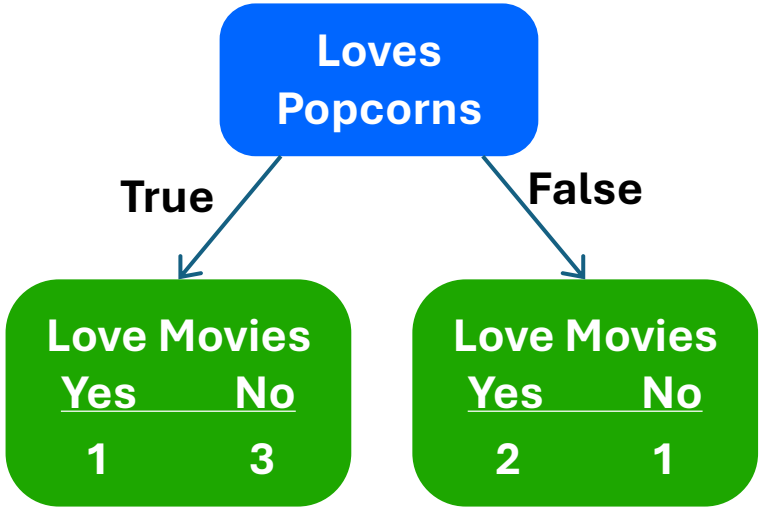
Classification: Decision Trees

The Working of Decision Trees?

Gini Impurity for Loves Popcorns= **0.405**

Gini Impurity for Loves Soda = **0.214**

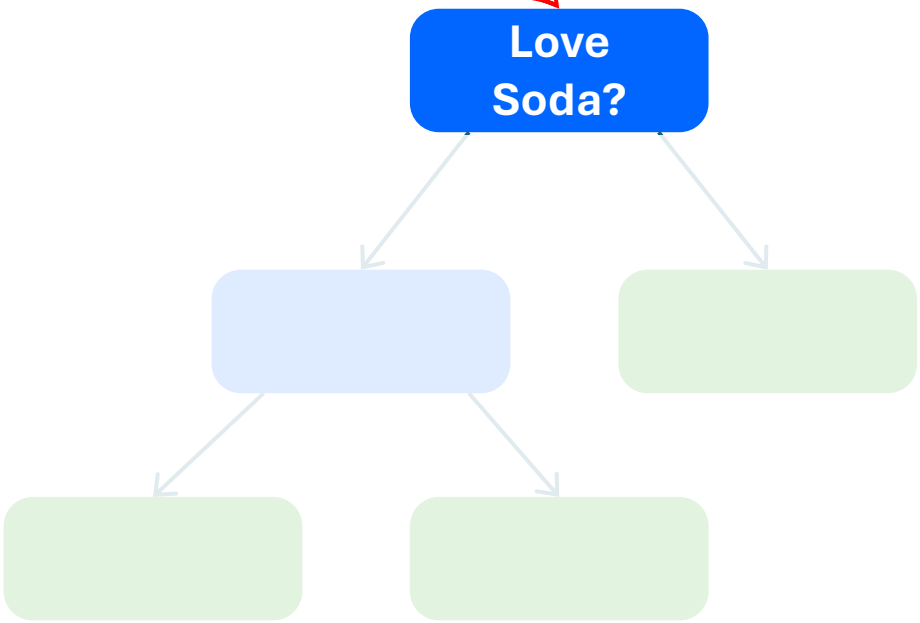
Gini Impurity for Age = **0.343**



Classification: Decision Trees

The Working of Decision Trees?

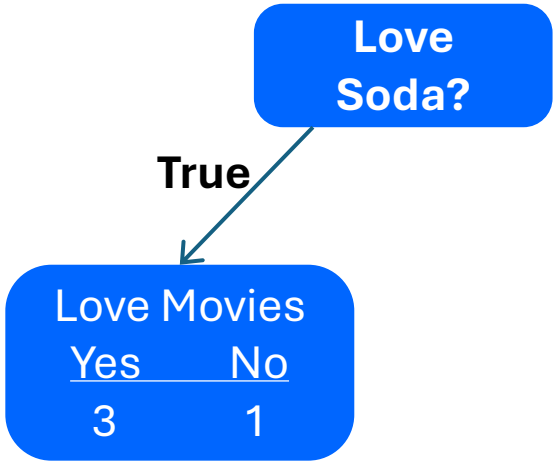
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Classification: Decision Trees

The Working of Decision Trees?

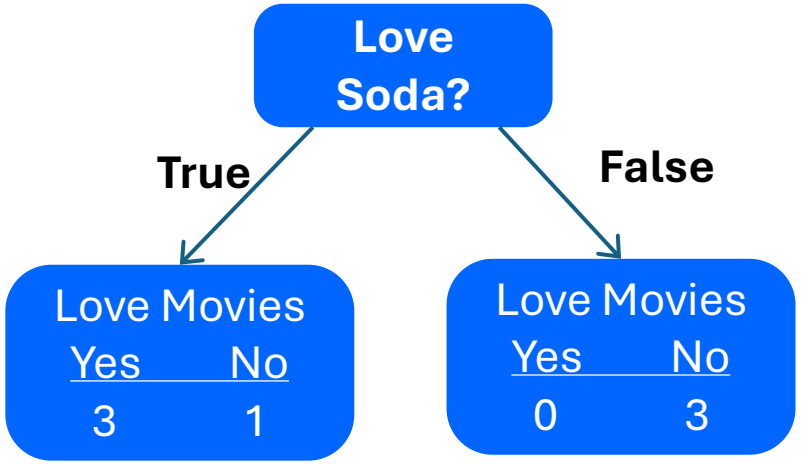
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Classification: Decision Trees

The Working of Decision Trees?

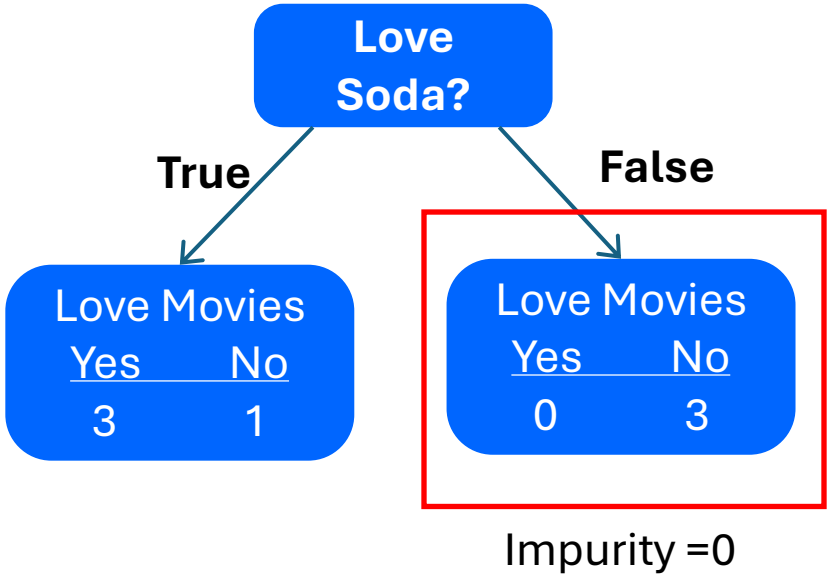
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Classification: Decision Trees

The Working of Decision Trees?

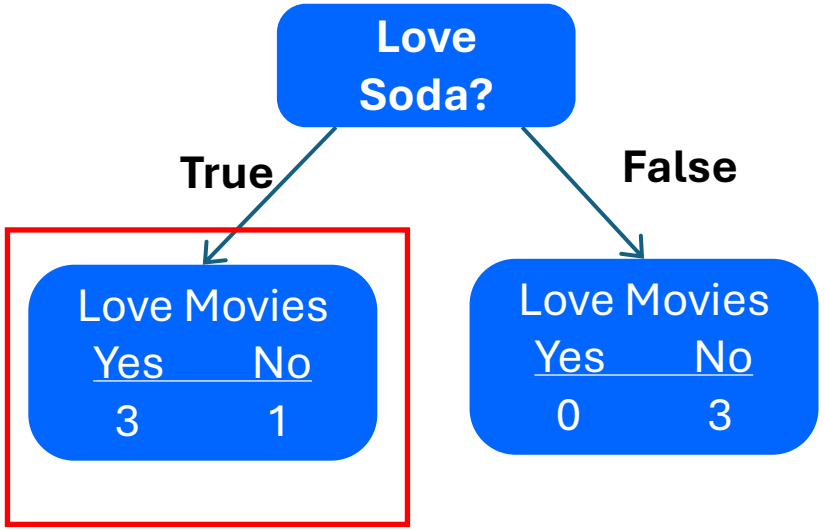
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

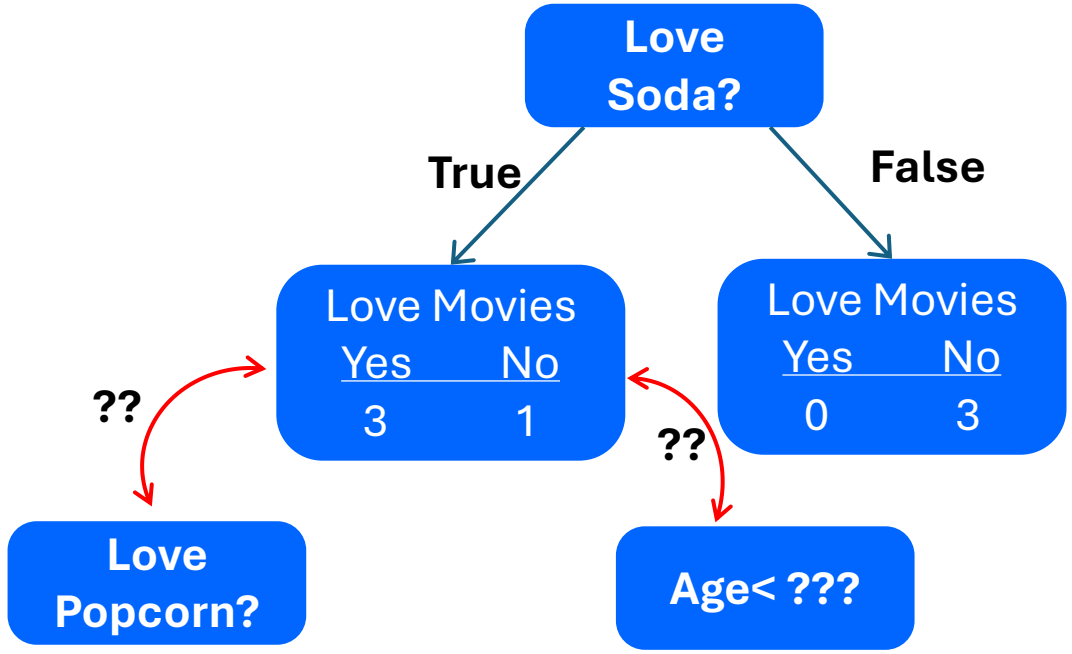


This node is Impure

Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



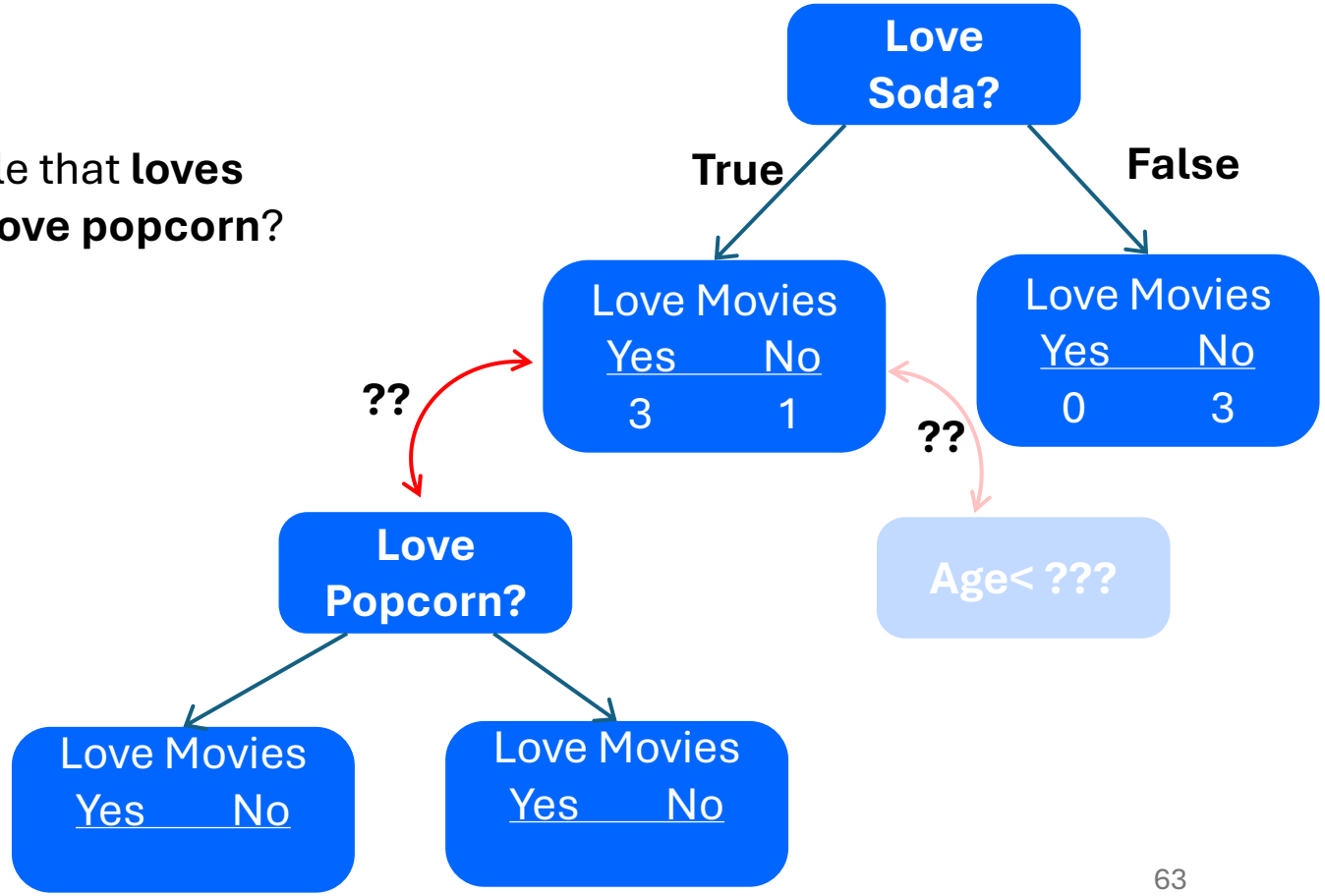
Lets reduce the impurity by slitting the people that love soda based on Loves Popcorn or Age.

Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

The 4 people that **loves soda** also **love popcorn**?

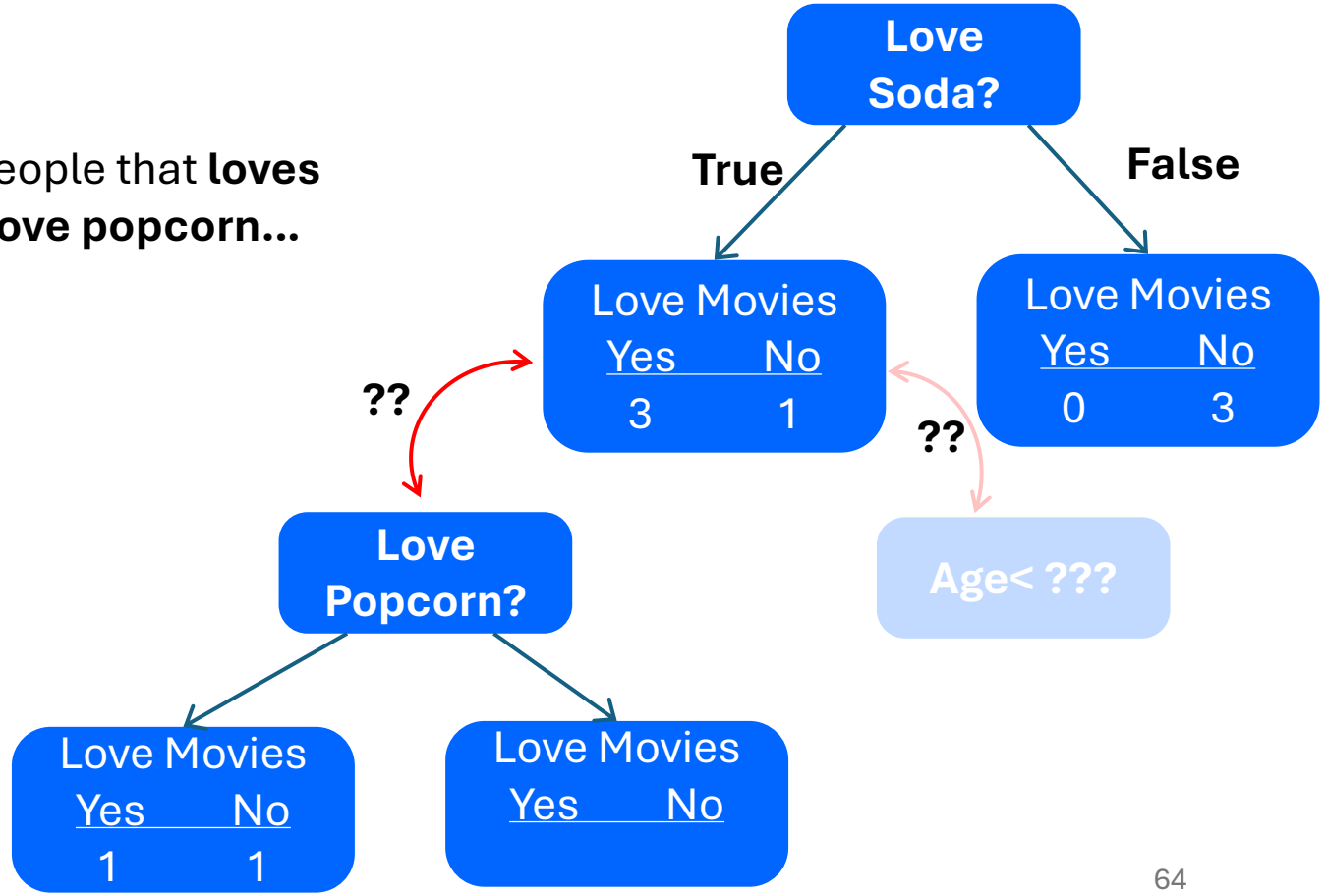


Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

2 of the 4 people that **loves soda** also **love popcorn…**

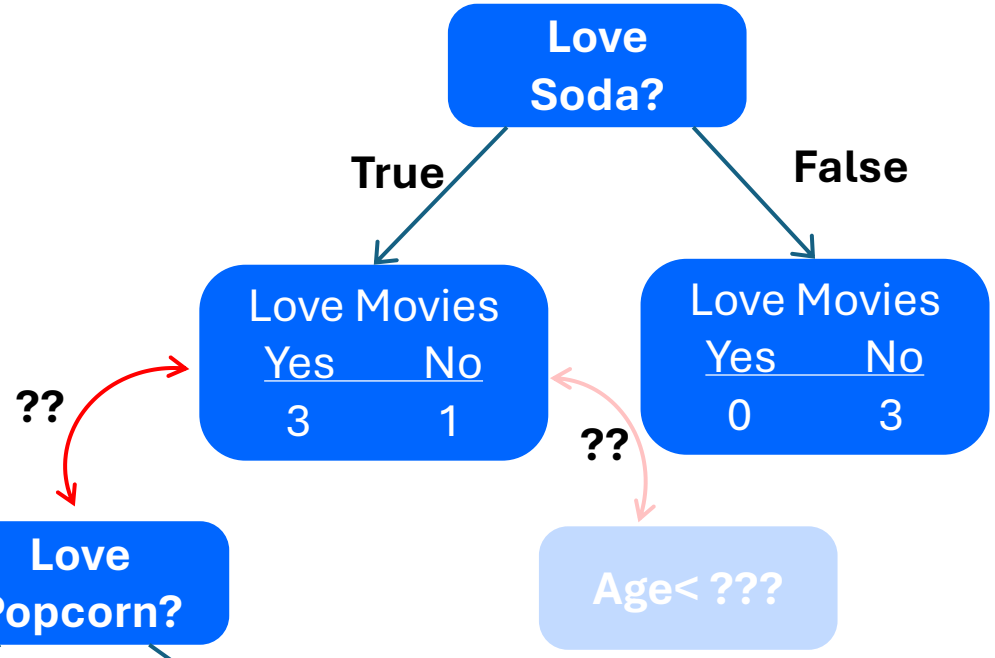


Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

And the remaining 2 people that **loves soda** but do not **love popcorn** end up in the right



Total impurity for this split = **0.25**

Classification: Decision Trees

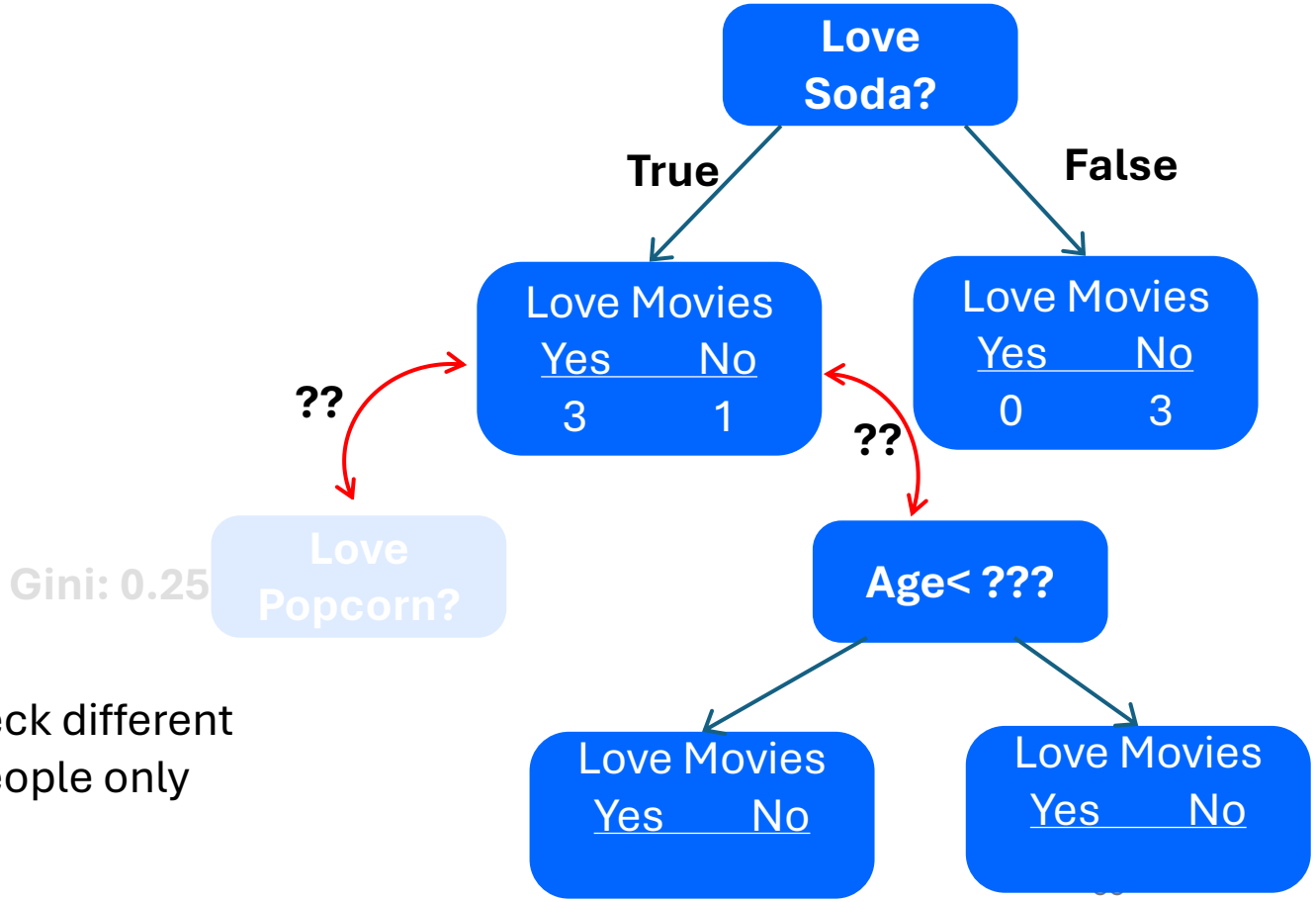
The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

12.5

26.5

36.5



This time we will check different age threshold, for people only **loves soda.**

Classification: Decision Trees

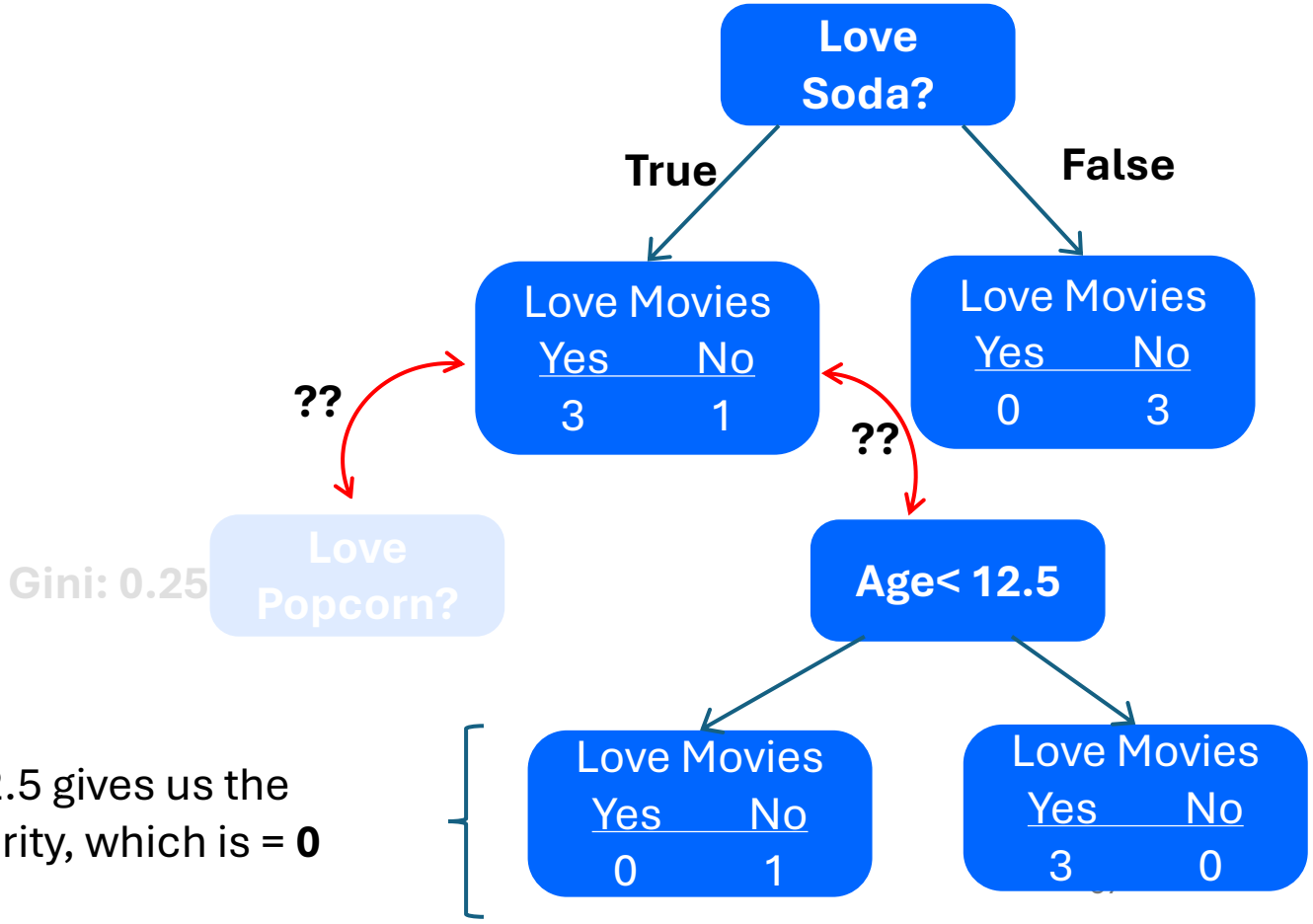
The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

12.5

26.5

36.5

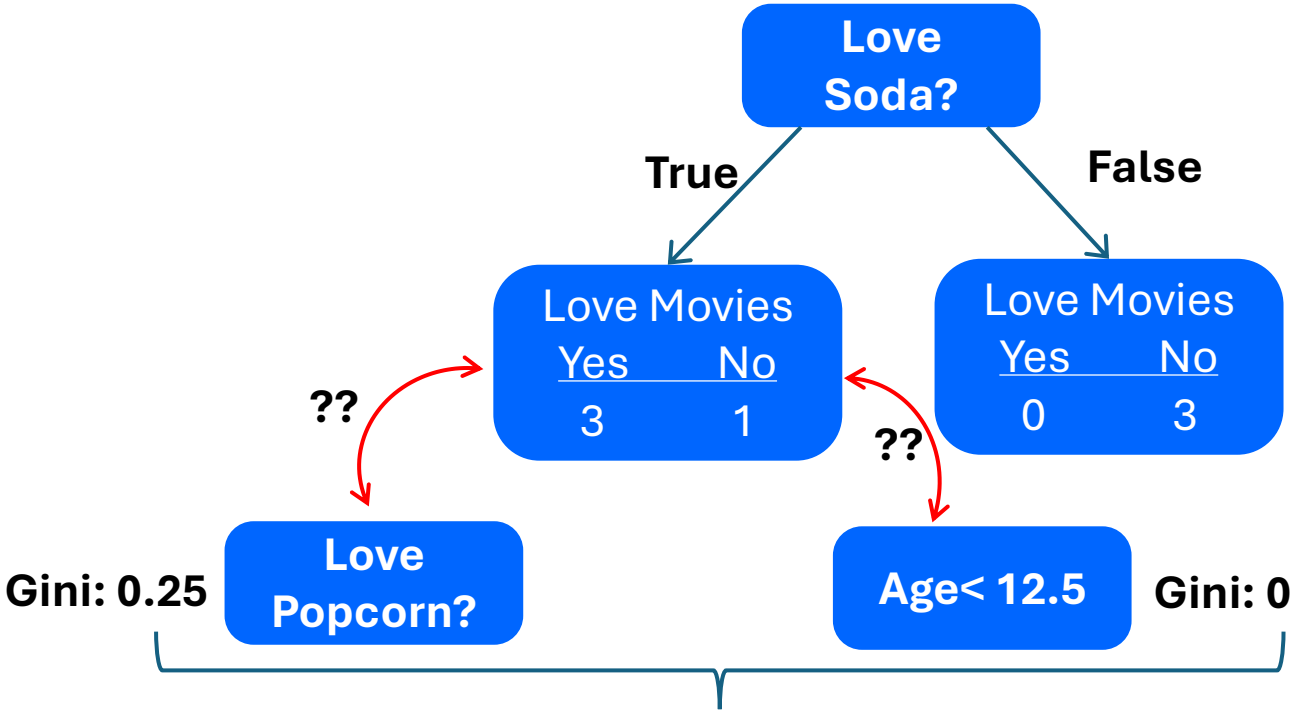


The age > 12.5 gives us the lowest impurity, which is = **0**

Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

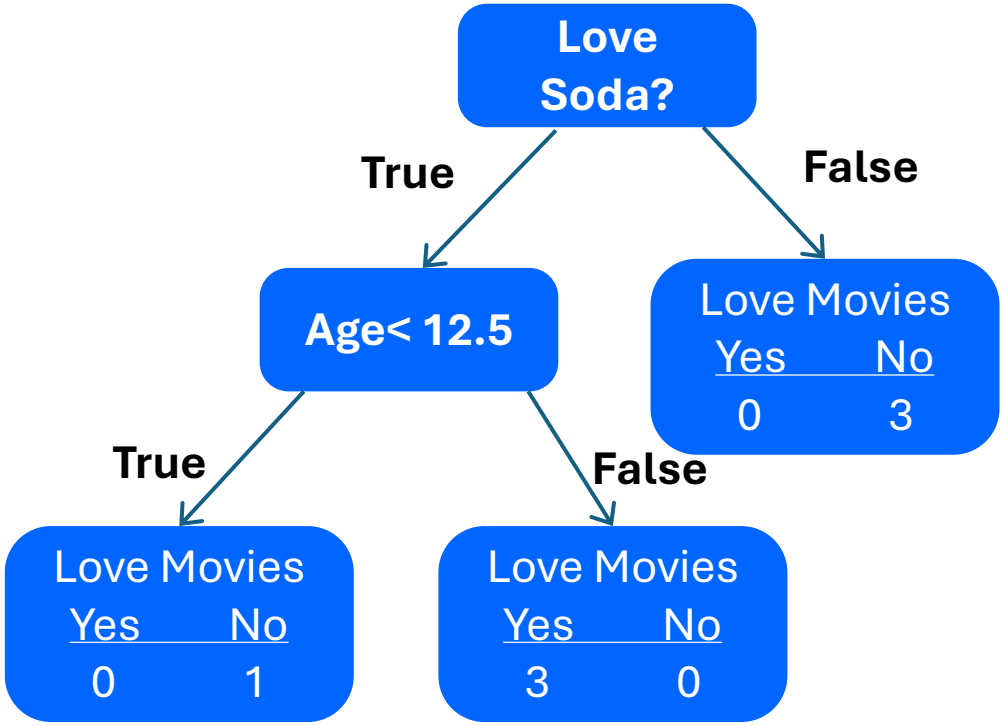


Now, **0** is less than **0.25,** so we will use **age<12.5** to split this node into leaves

Classification: Decision Trees

The Working of Decision Trees?

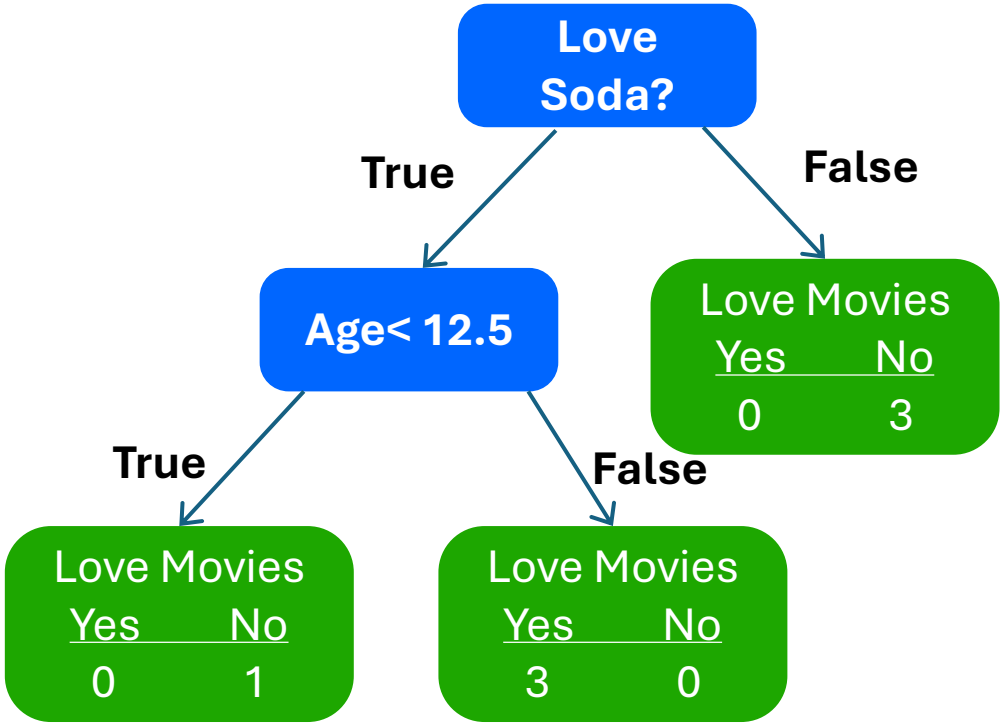
Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



Note: these are leaves because there is no reason to continue splitting these people into smaller groups.

Classification: Decision Trees

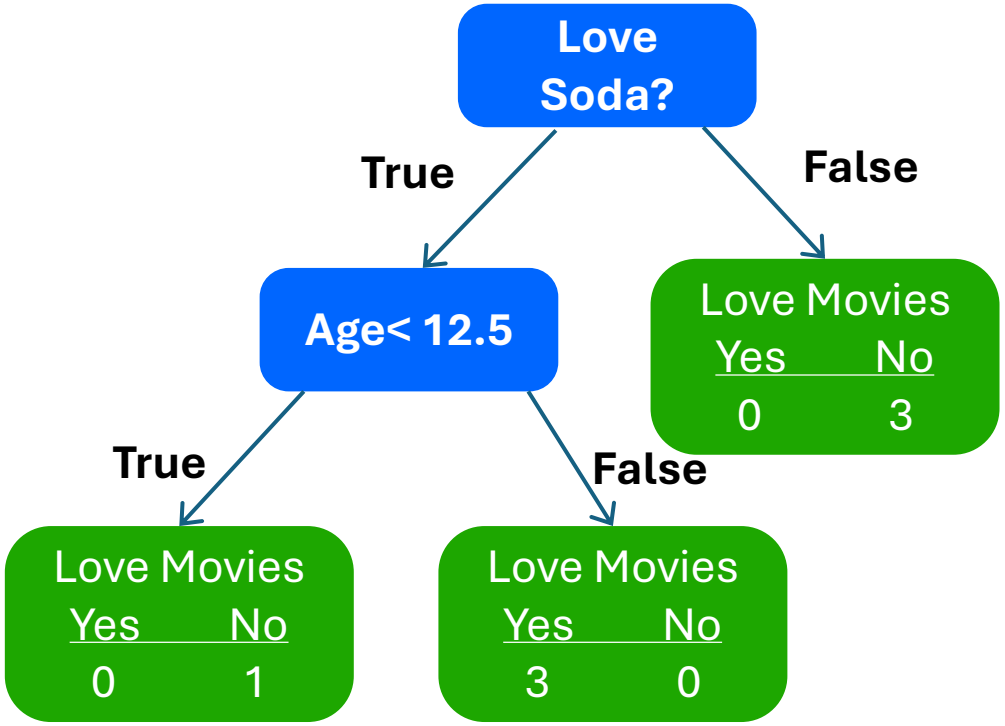
The Working of Decision Trees?



Note: these are leaves because there is no reason to continue splitting these people into smaller groups.

Classification: Decision Trees

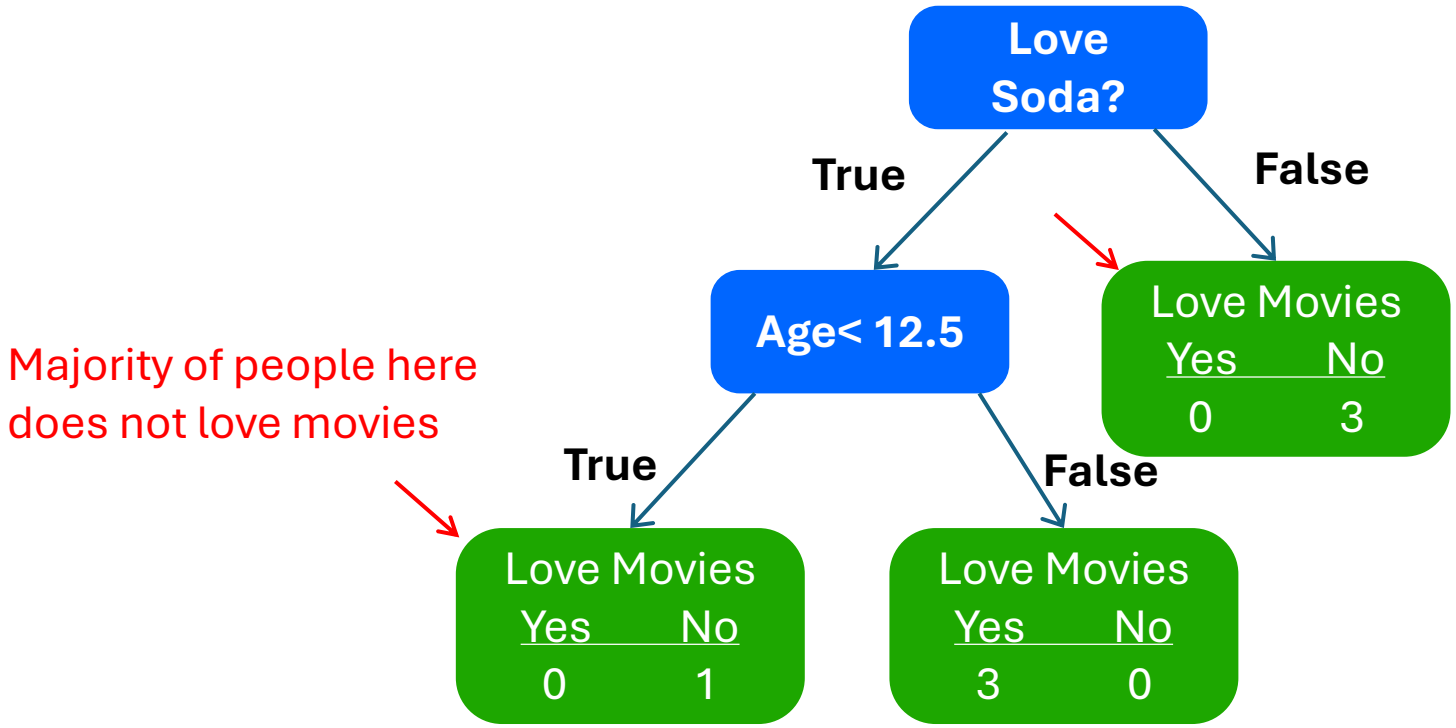
The Working of Decision Trees?



Final step: now we need to assign output values/labels for each leaf…
The output of a leaf is whatever category that has the most votes

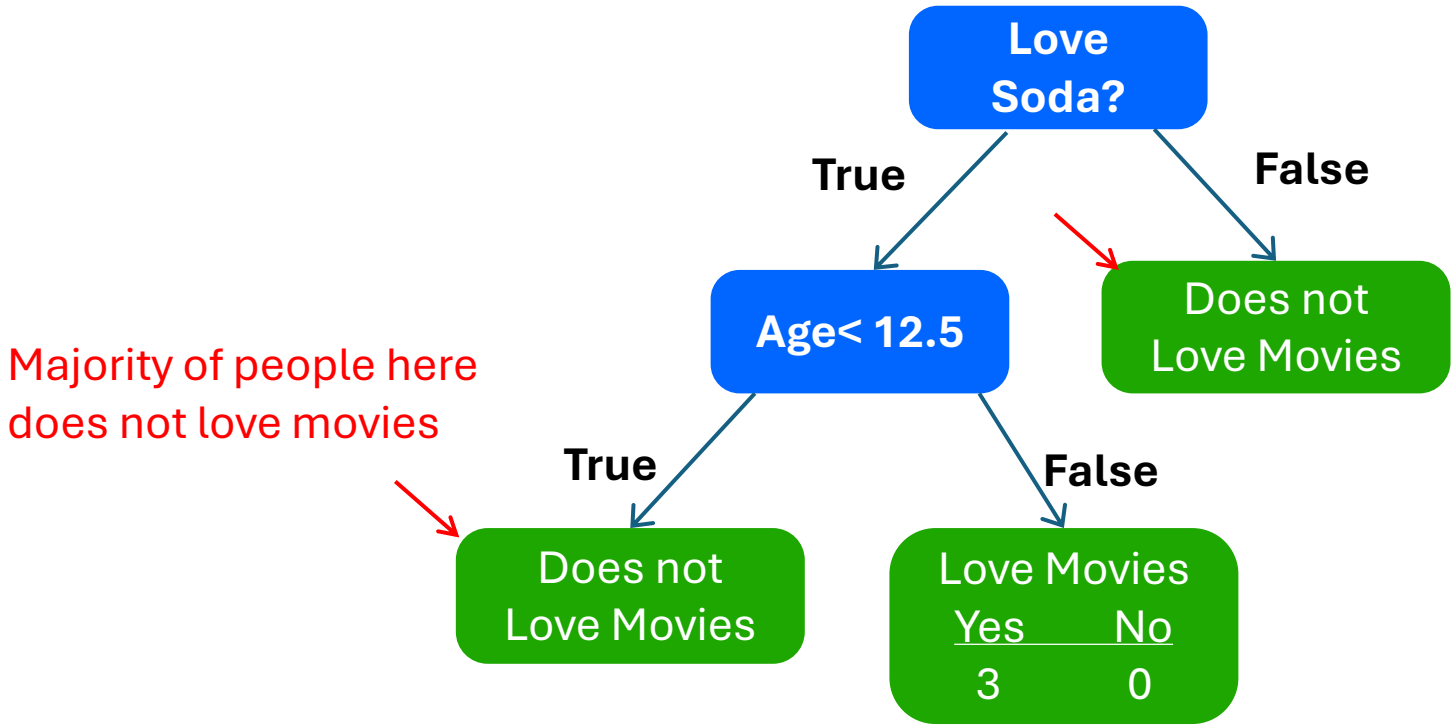
Classification: Decision Trees

The Working of Decision Trees?



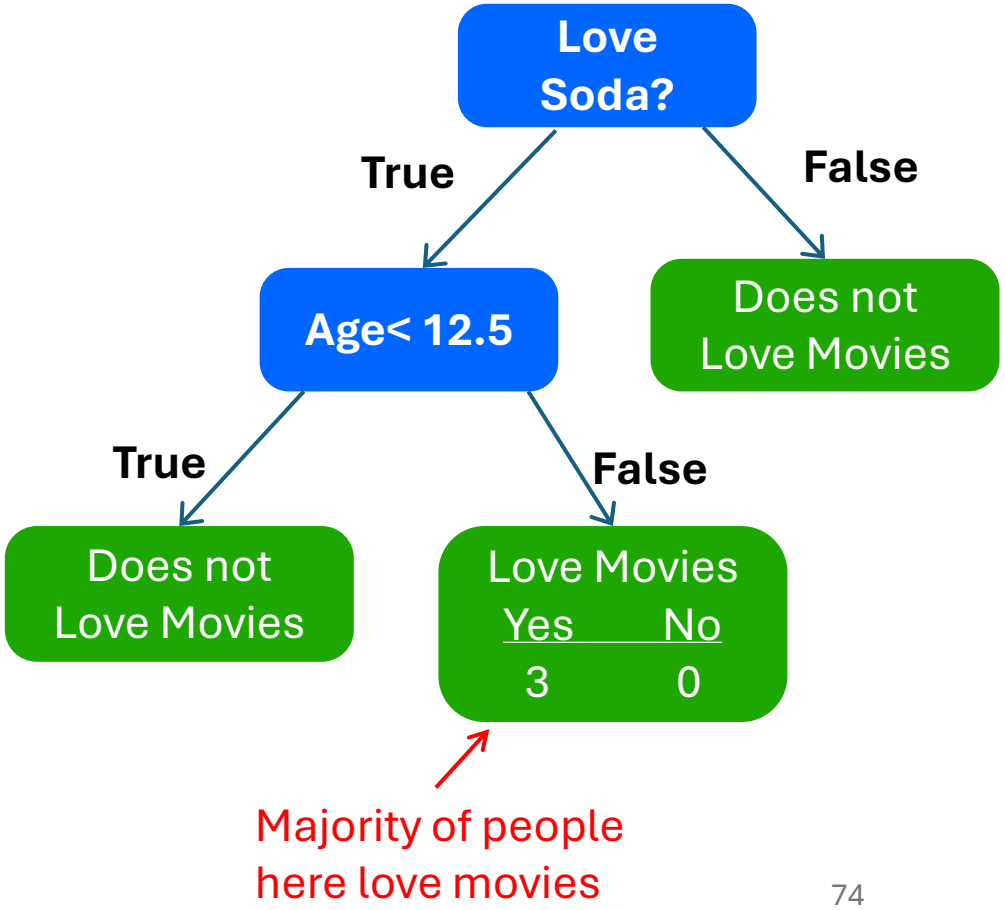
Classification: Decision Trees

The Working of Decision Trees?



Classification: Decision Trees

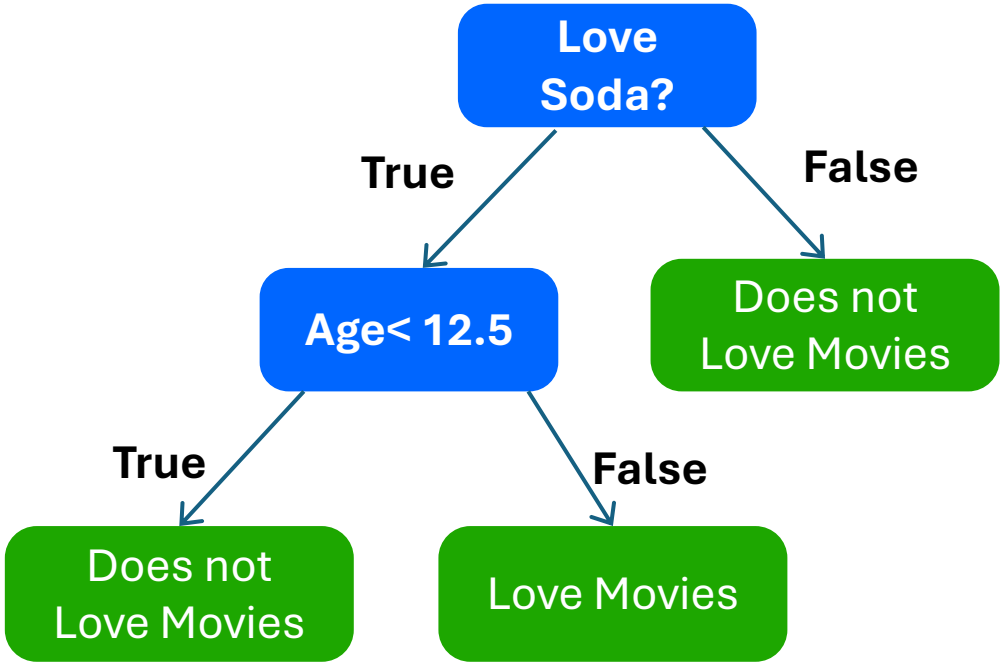
The Working of Decision Trees?



Classification: Decision Trees

The Working of Decision Trees?

Finally, we completed the tree from the data.

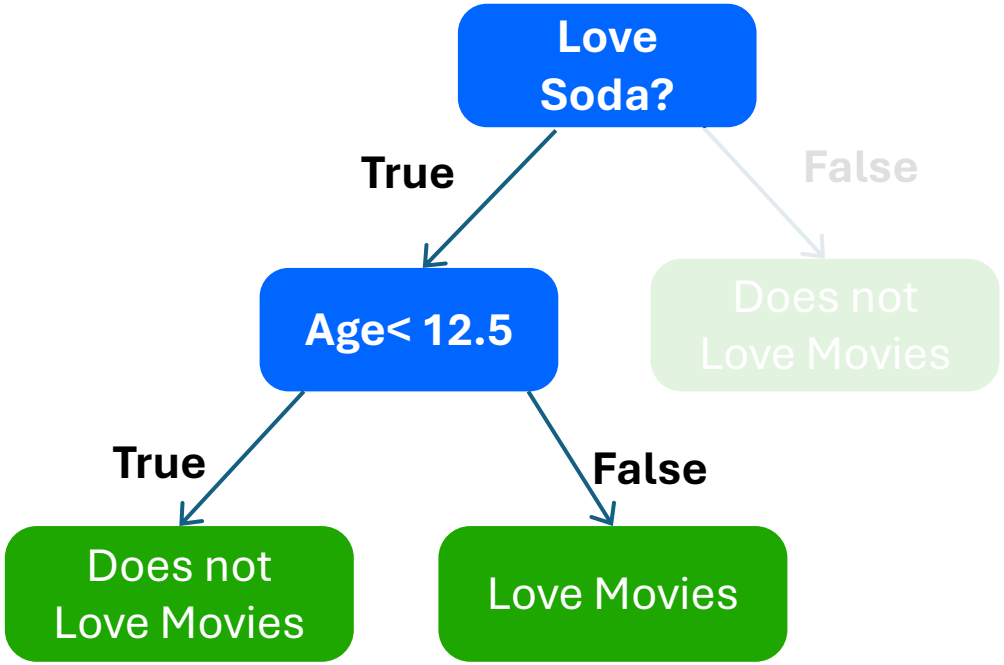


Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	15	??

Finally, we completed the tree from the data.

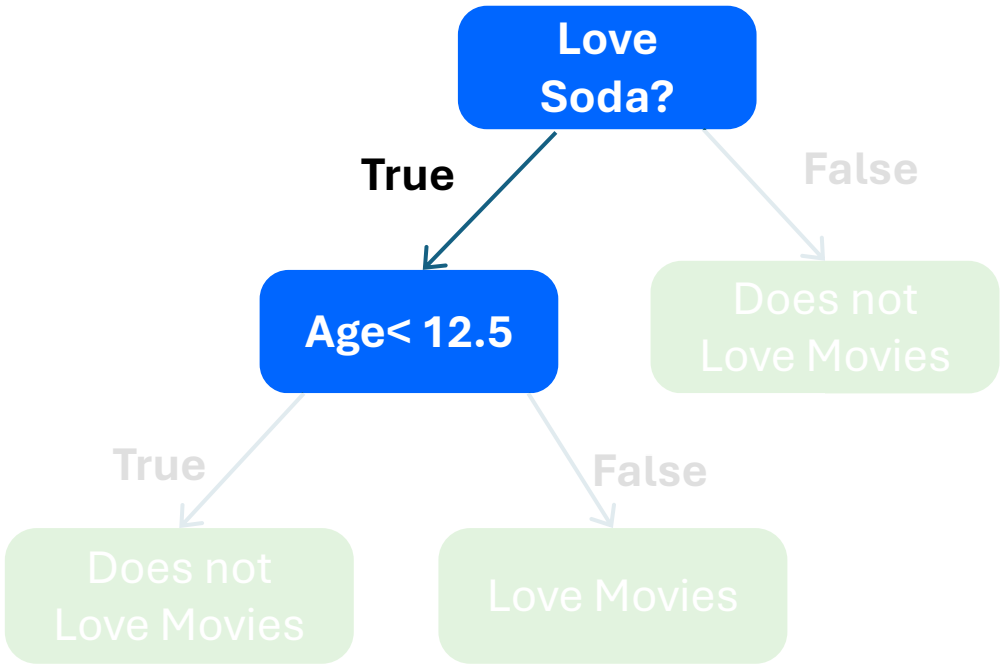


Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	15	??

Finally, we completed the tree from the data.

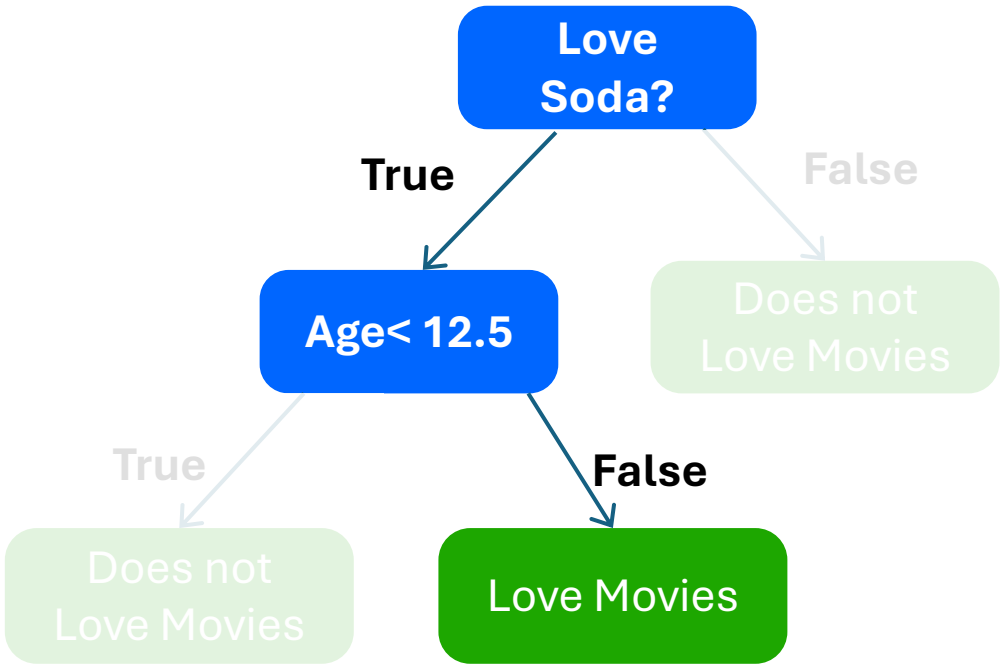


Classification: Decision Trees

The Working of Decision Trees?

Loves Popcorns	Loves Soda	Age	Love Movies
Yes	Yes	15	??

Finally, we completed the tree from the data.



Classification: Decision Trees

Our Question For Today?

The scenario

- A property assessor must place every house in a price band before the valuation is finalized.
- They have the full record: size, quality, year built, garage, everything.
- They do not yet have the price.

The question?

Given everything else about the house, is it **above** or **below** the market median?

Why this question works?

- The answer is a category, not a number. That makes it classification.
- Two classes, roughly half the houses in each. Balanced.
- A real decision. Somebody acts differently depending on the answer.